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(54) **IMAGE SYSTEM LENS ASSEMBLY**

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G02B 3/02 (2006.01)

(52) **U.S. Cl.**
CPC **G02B 9/62** (2013.01)
USPC **359/761; 359/713**

(58) **Field of Classification Search**

USPC 359/713, 756, 761
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

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* cited by examiner

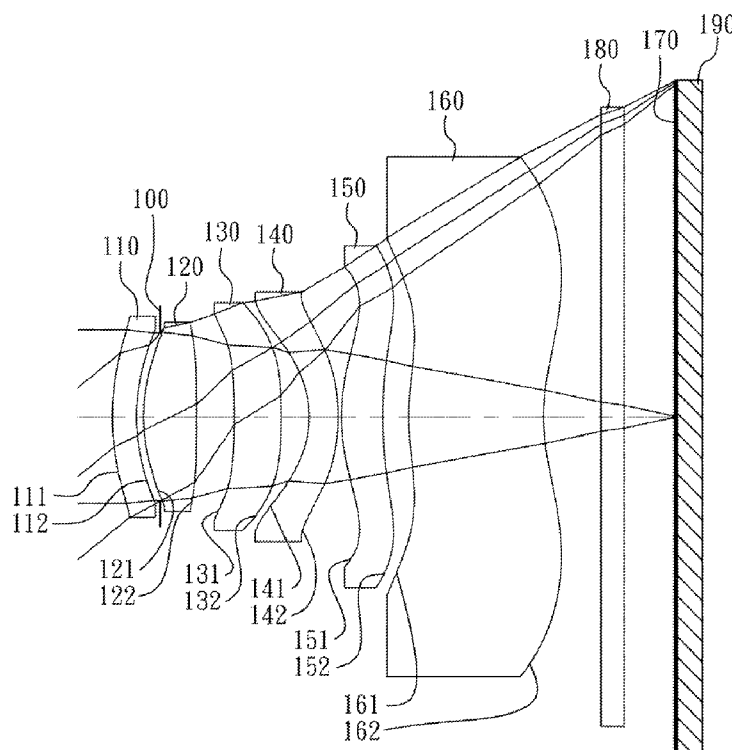
Primary Examiner — Jack Dinh

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(57) **ABSTRACT**

An image system lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element with negative refractive power has a convex object-side surface and a concave image-side surface. The second lens element has positive refractive power. The third lens element has refractive power. The fourth lens element with negative refractive power has a concave object-side surface and a convex image-side surface. The plastic fifth lens element with positive refractive power has at least one aspheric surface. The plastic sixth lens element with refractive power has a convex object-side surface and a concave image-side surface, and has at least one aspheric surface, wherein the image-side surface thereof changes from concave at a paraxial region to convex at a peripheral region.

29 Claims, 20 Drawing Sheets



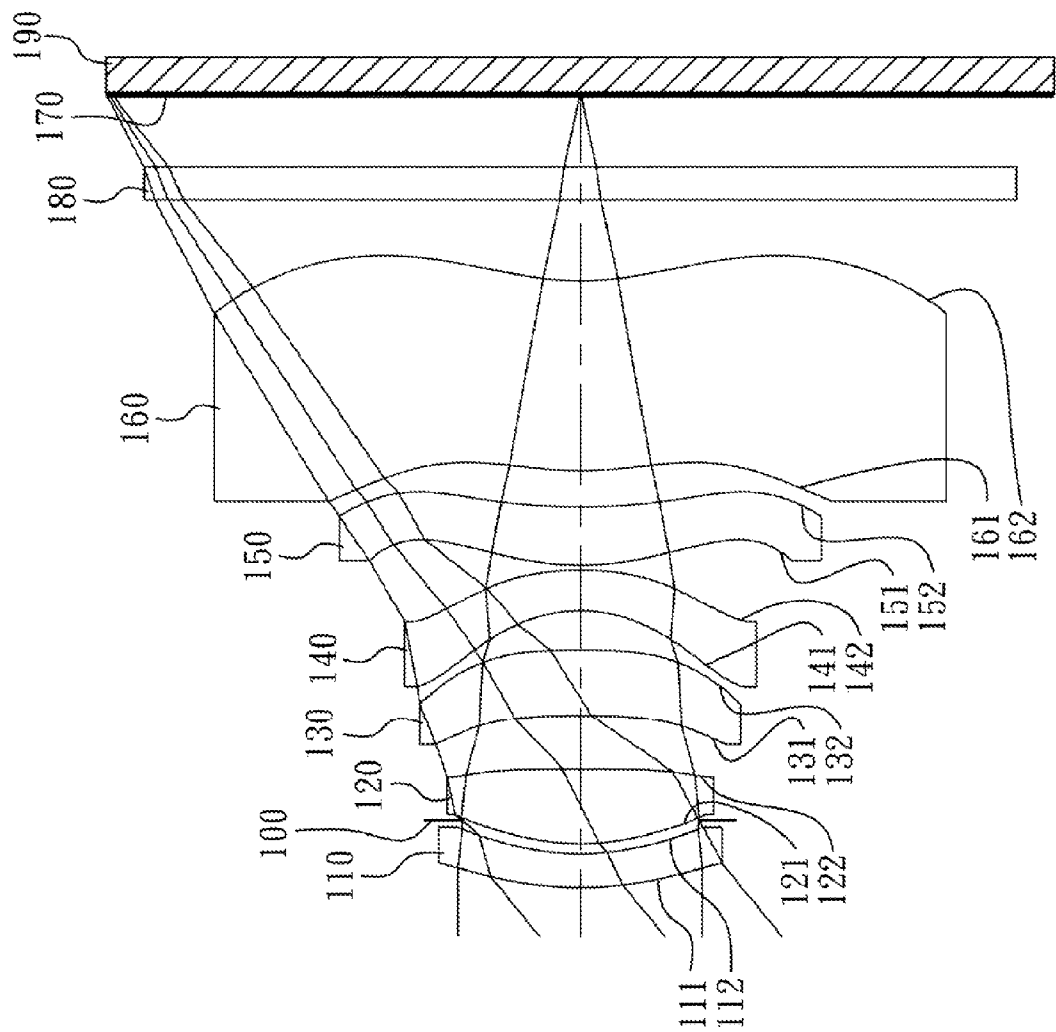


Fig. 1

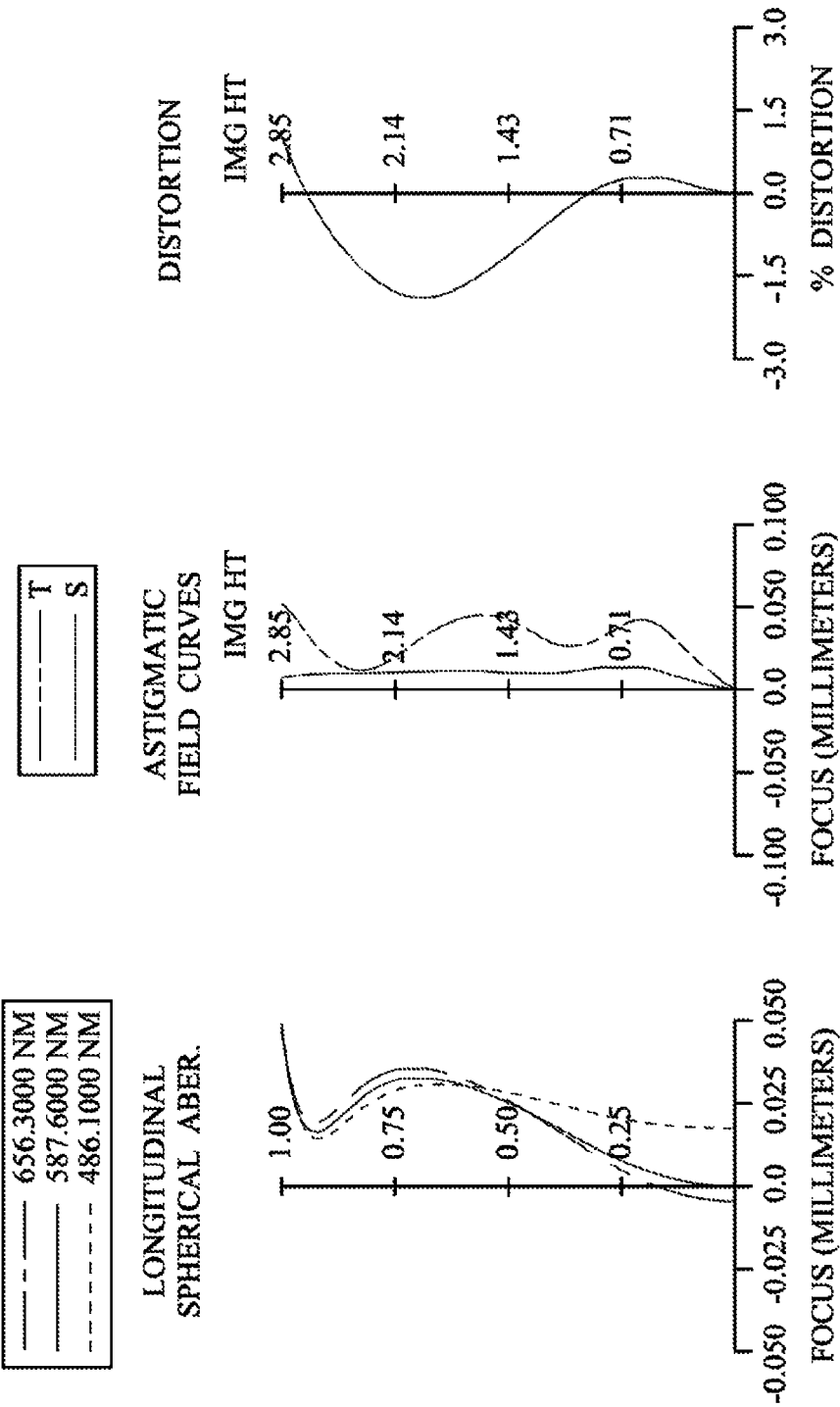


Fig. 2

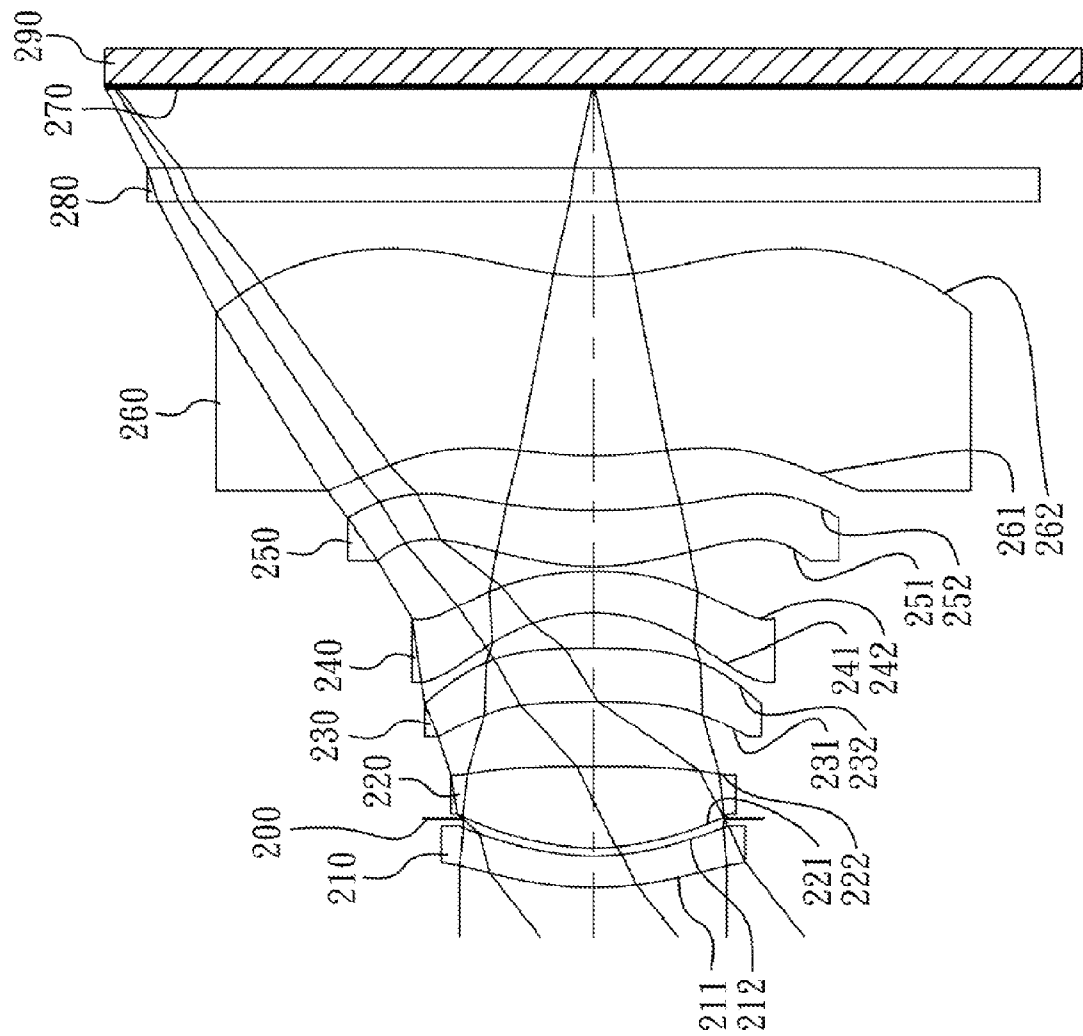


Fig. 3

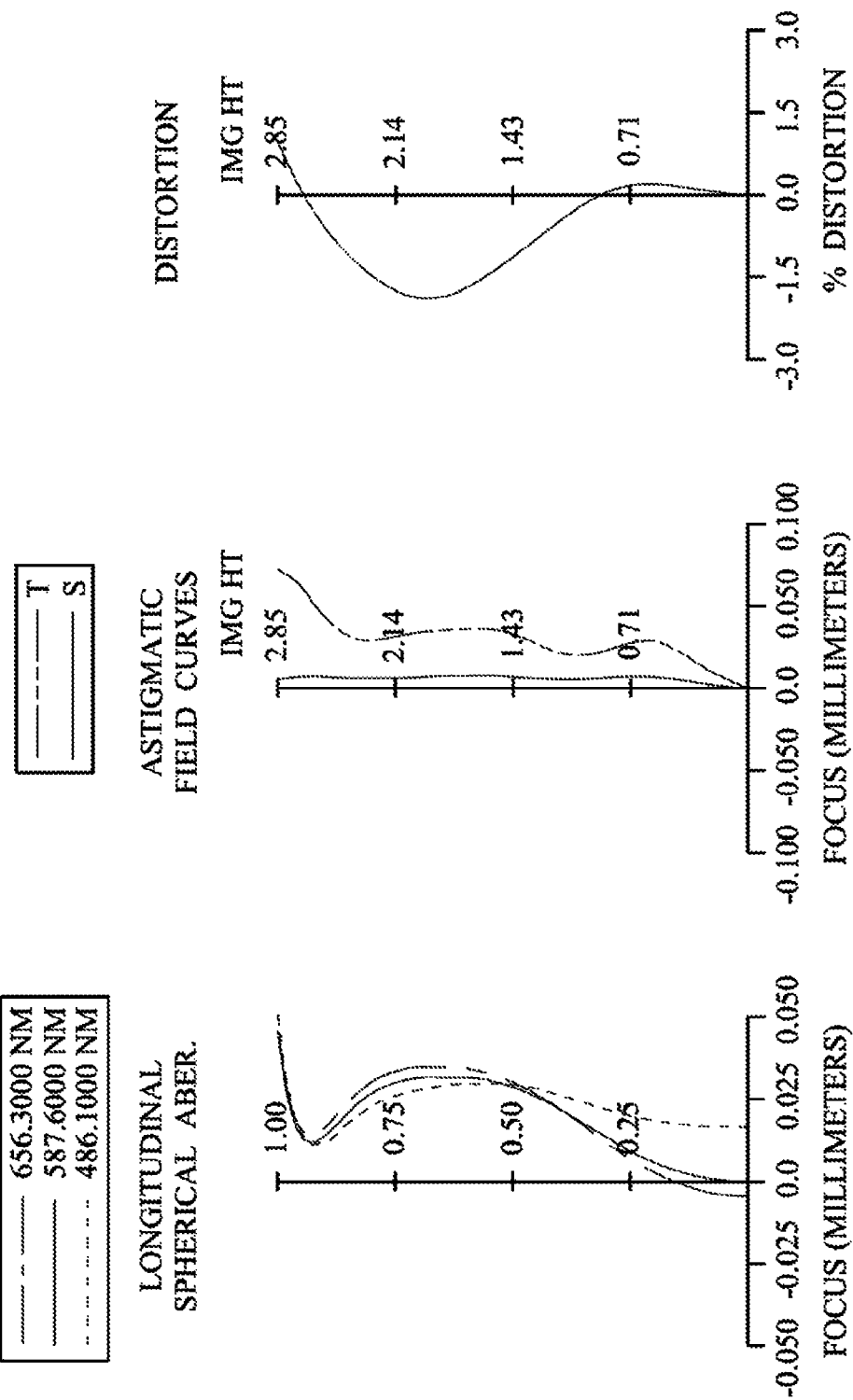


Fig. 4

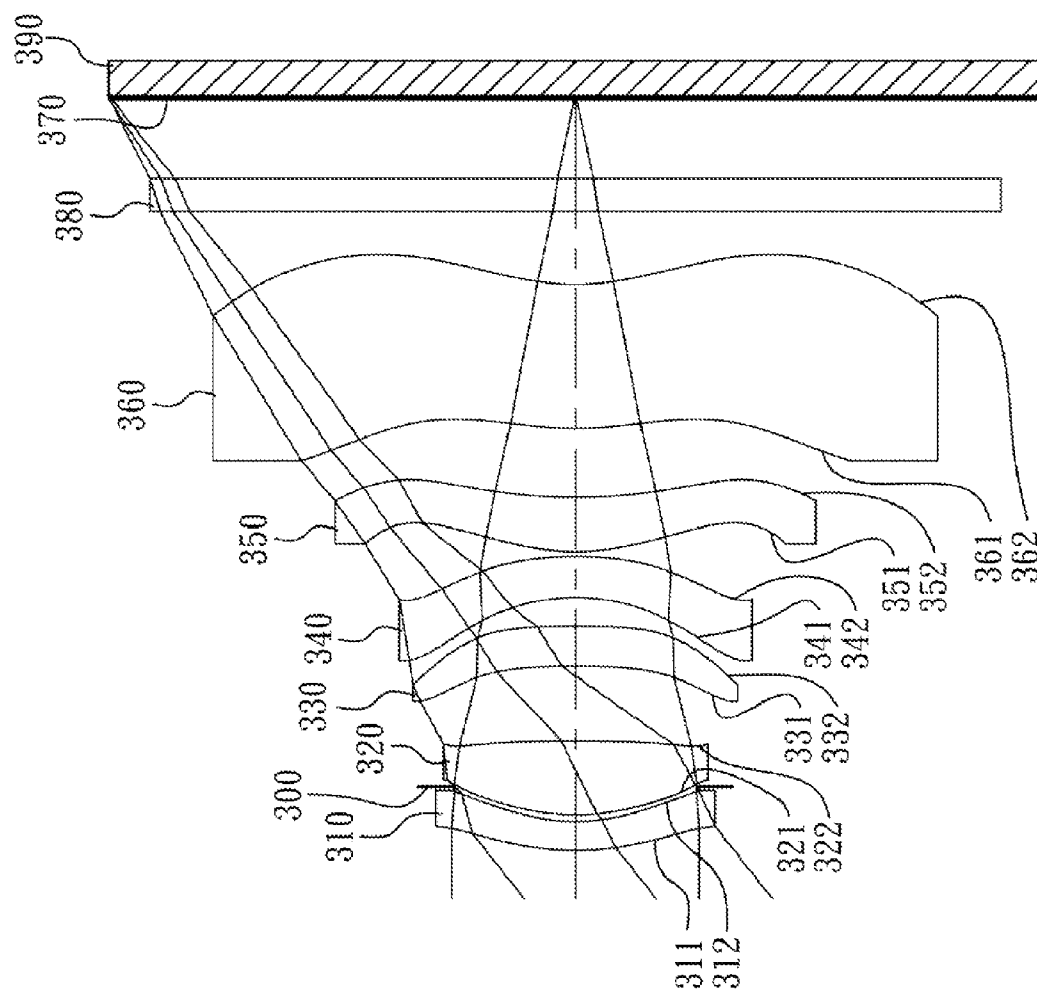


Fig. 5

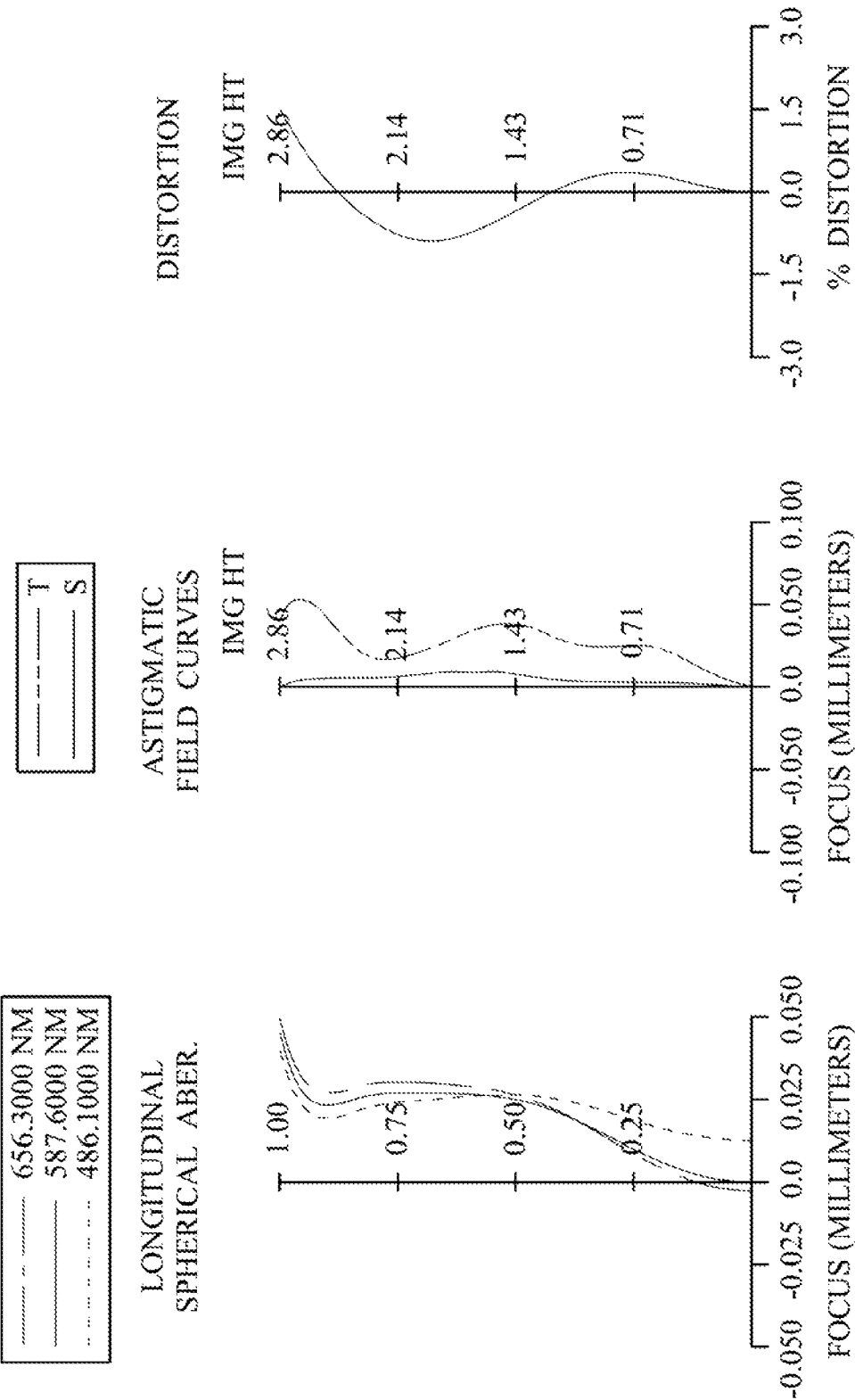


Fig. 6

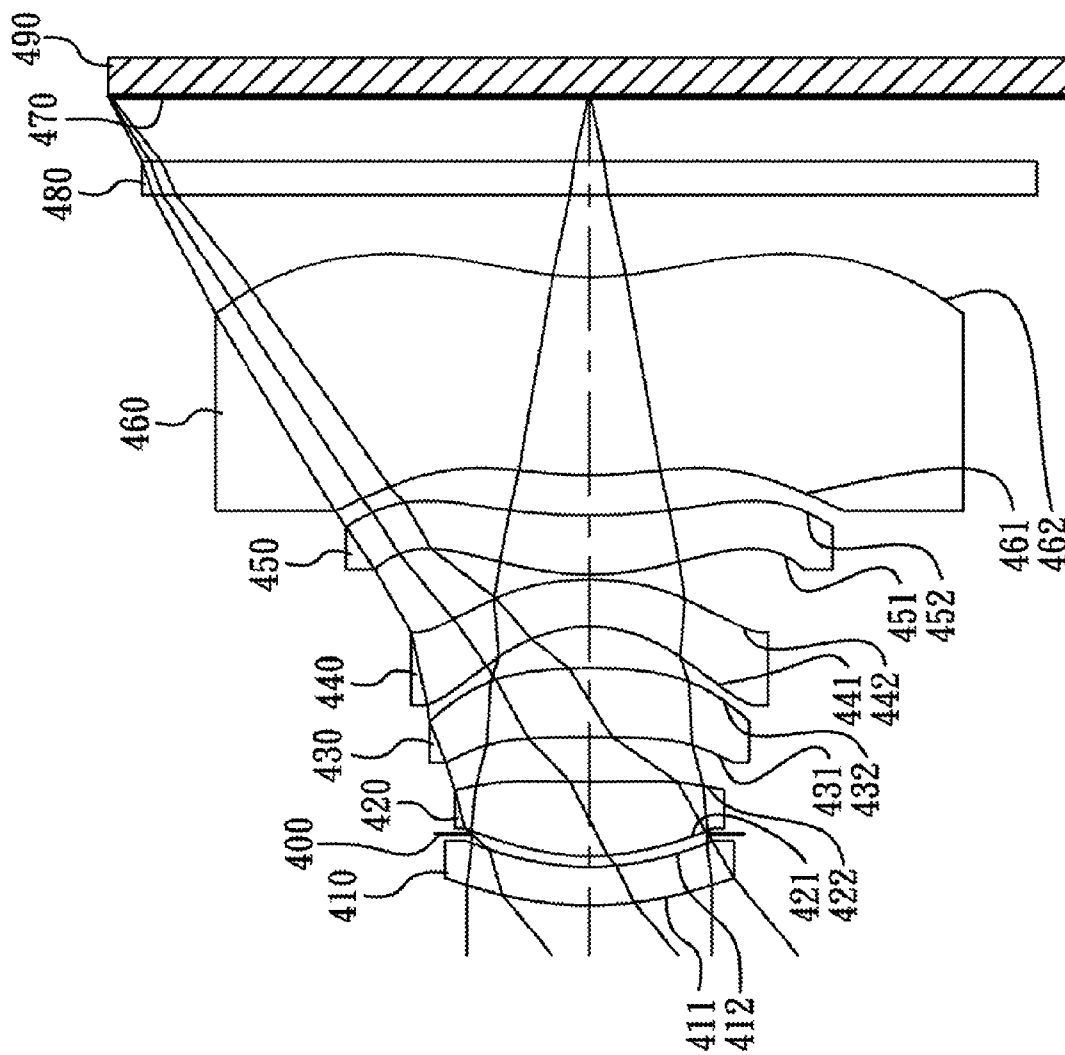


Fig. 7

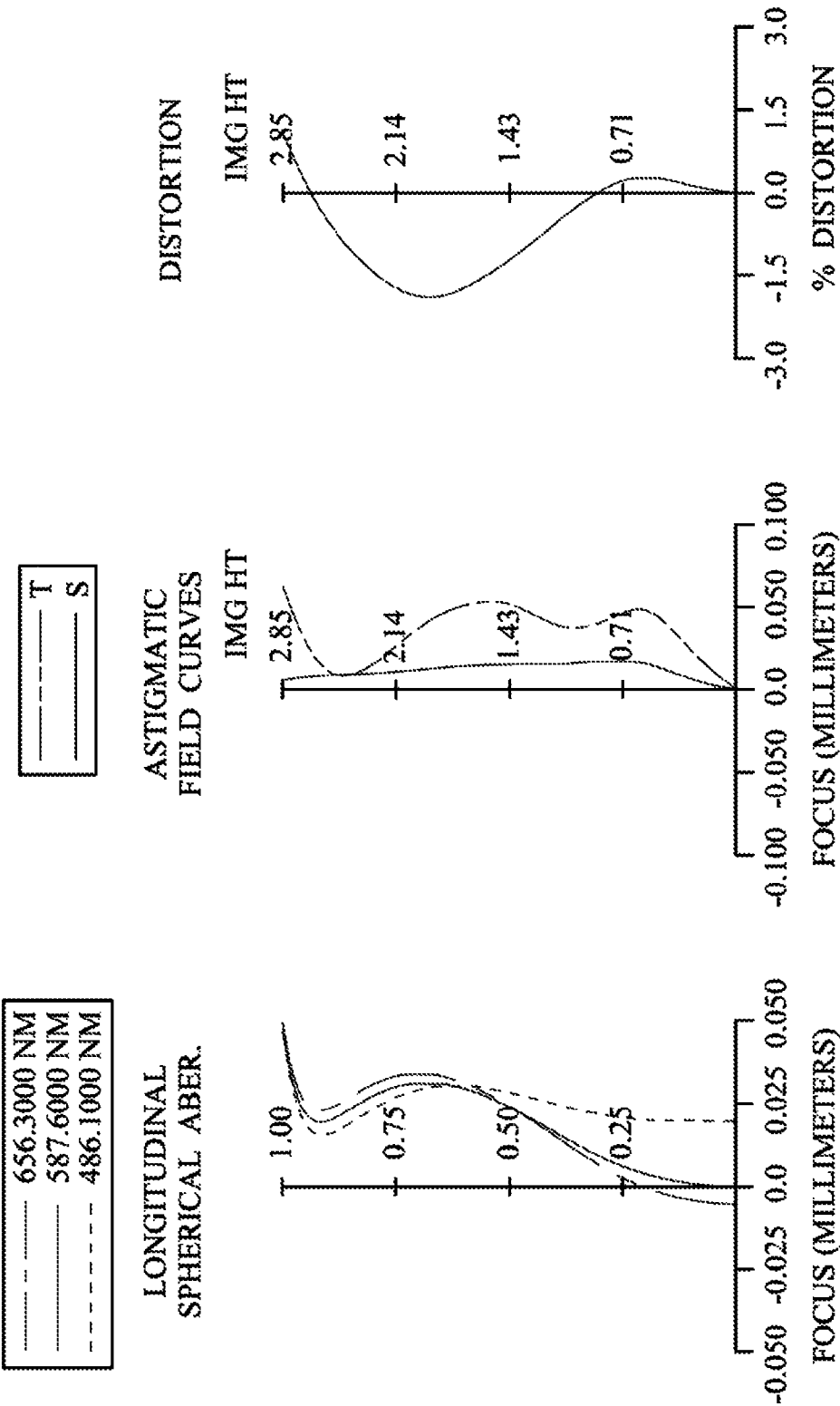


Fig. 8

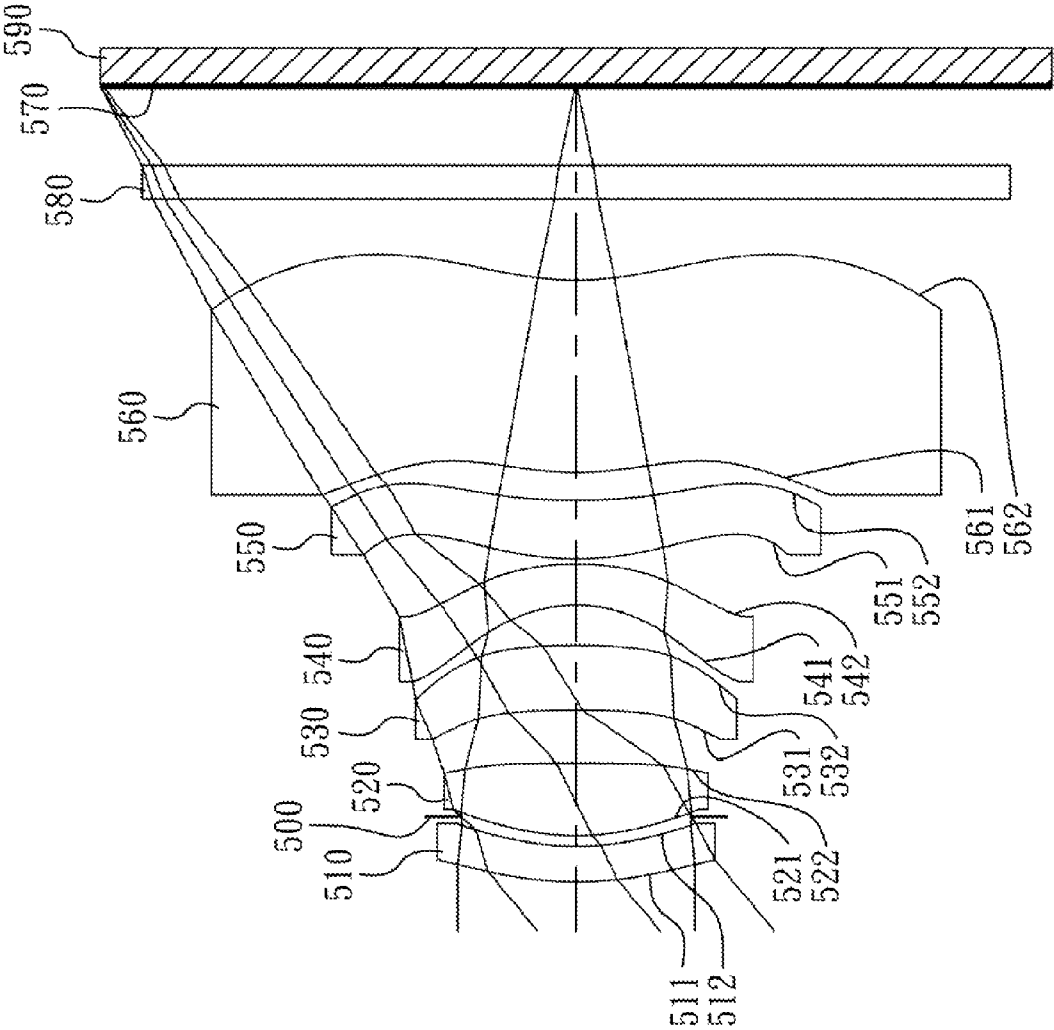


Fig. 9

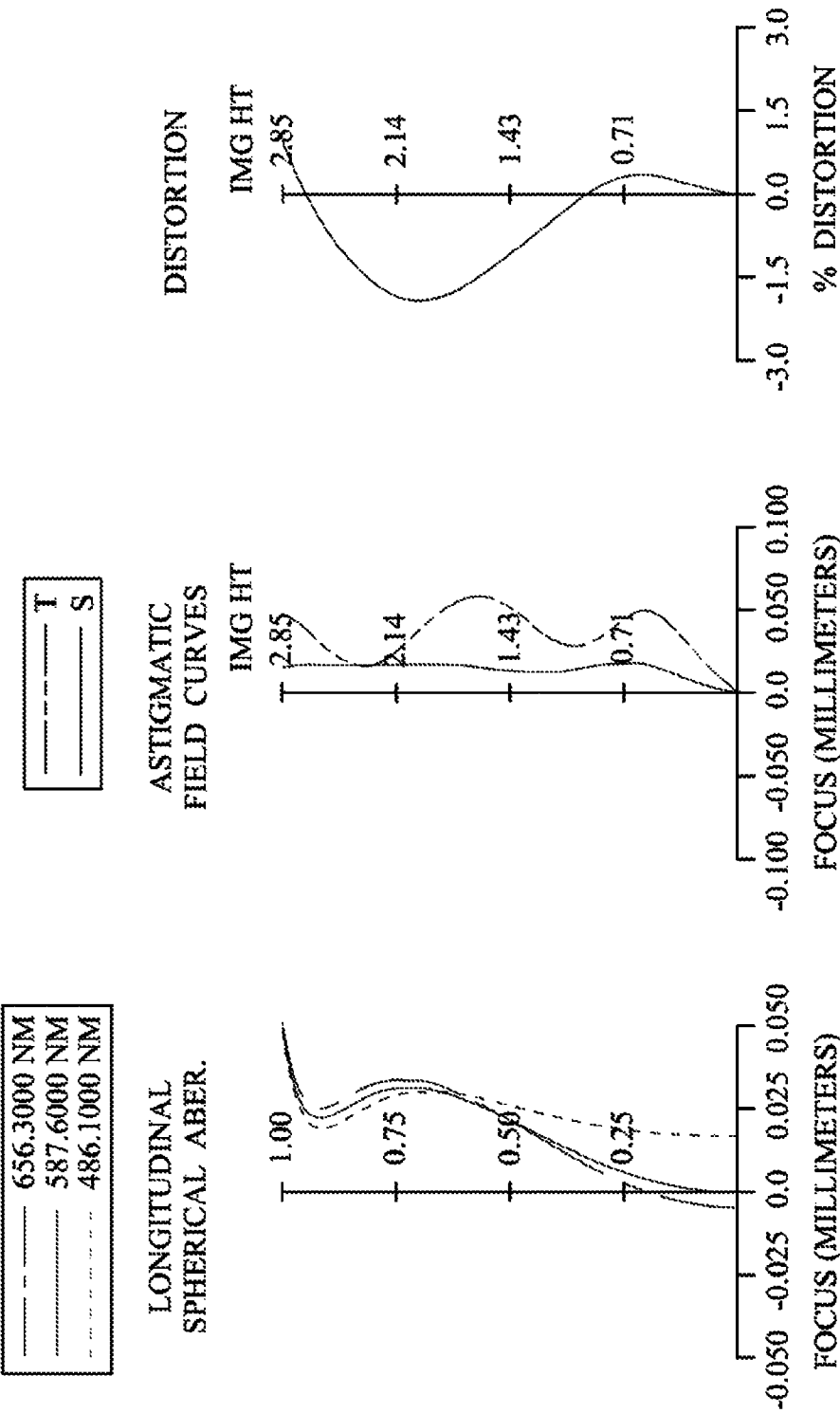


Fig. 10

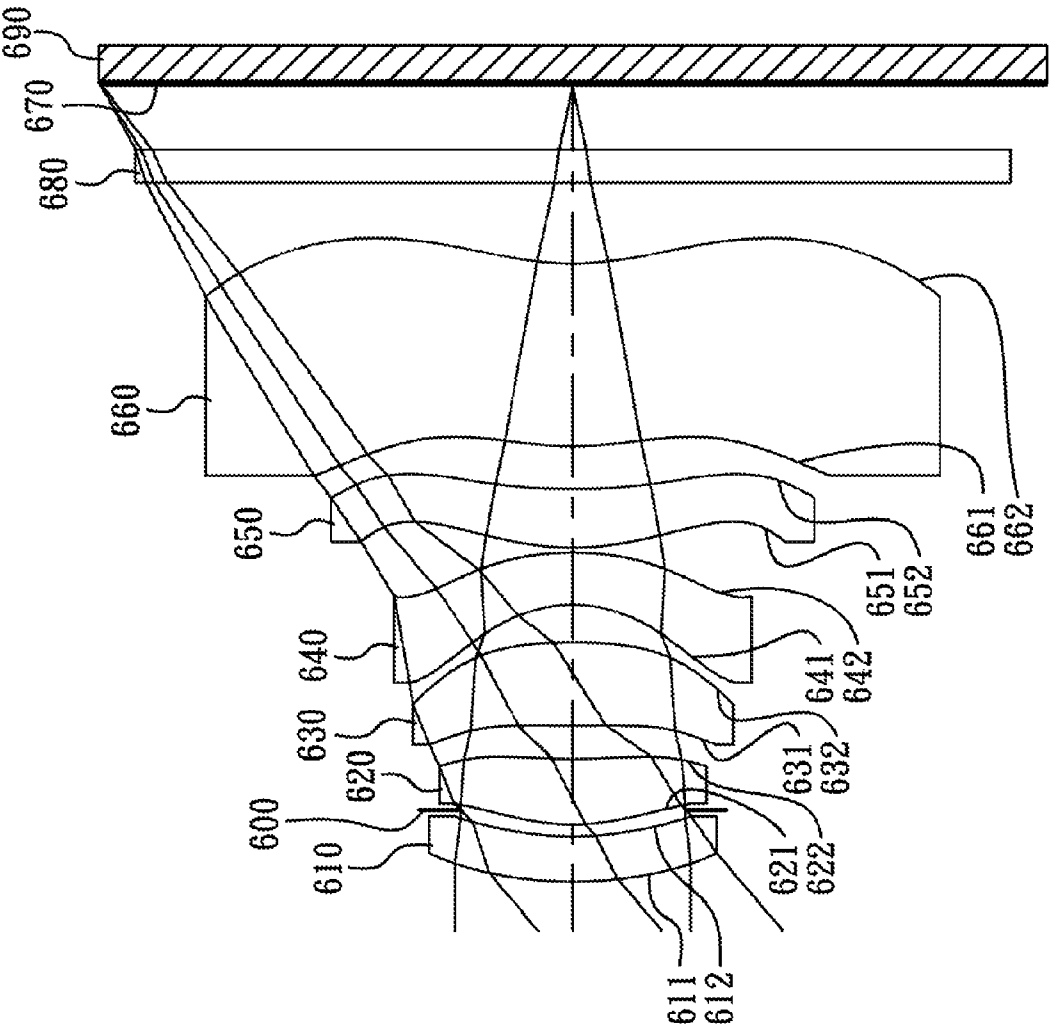


Fig. 11

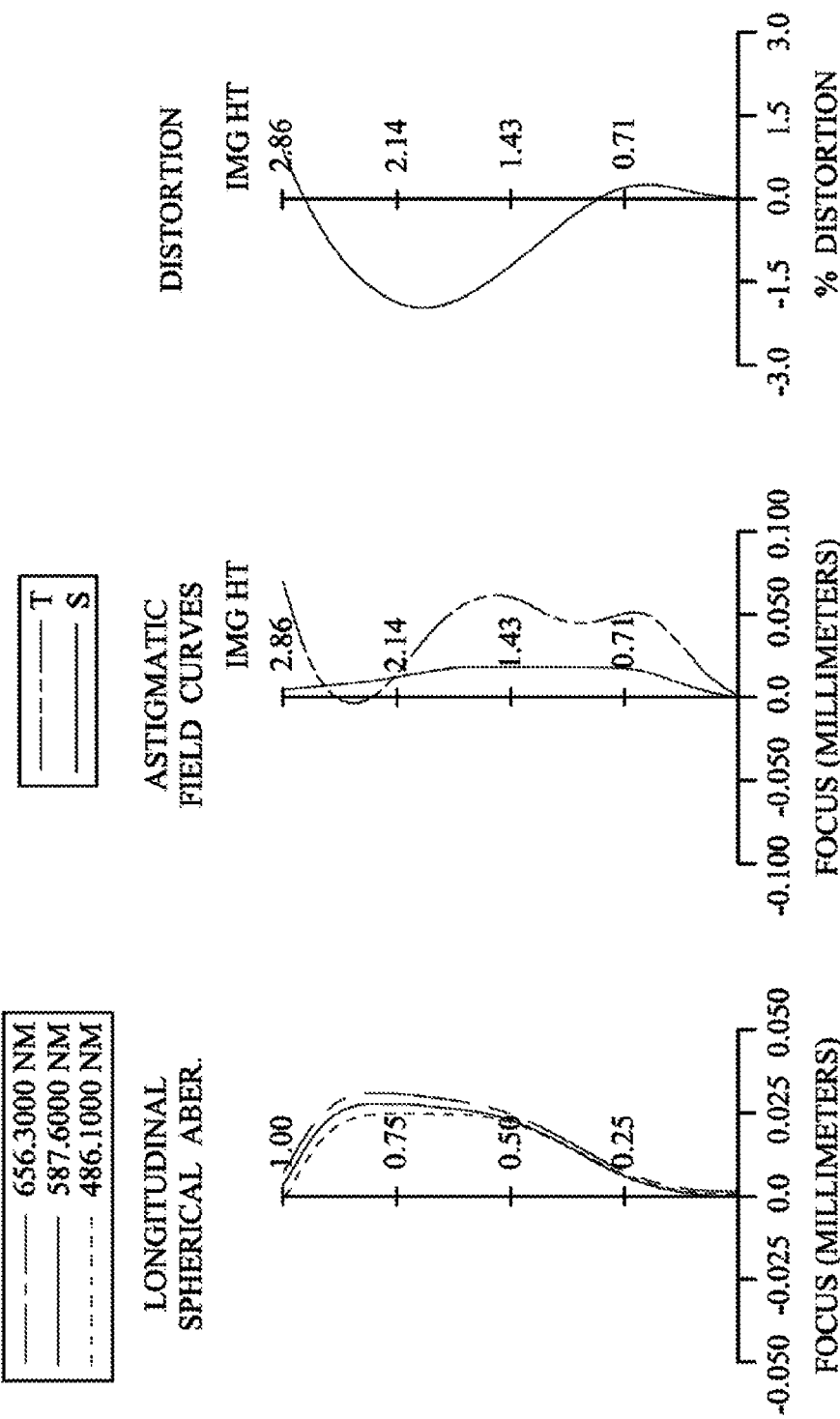


Fig. 12

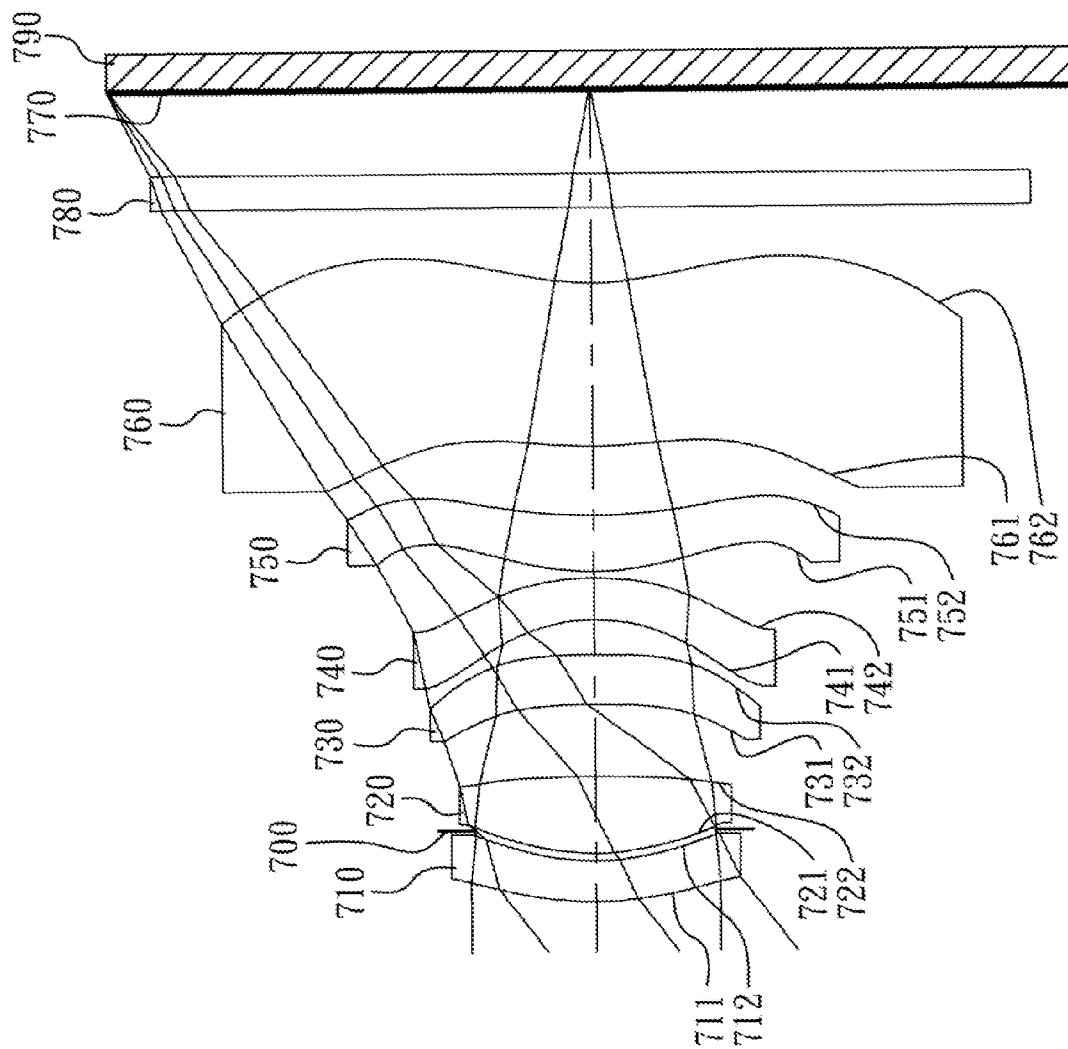


Fig. 13

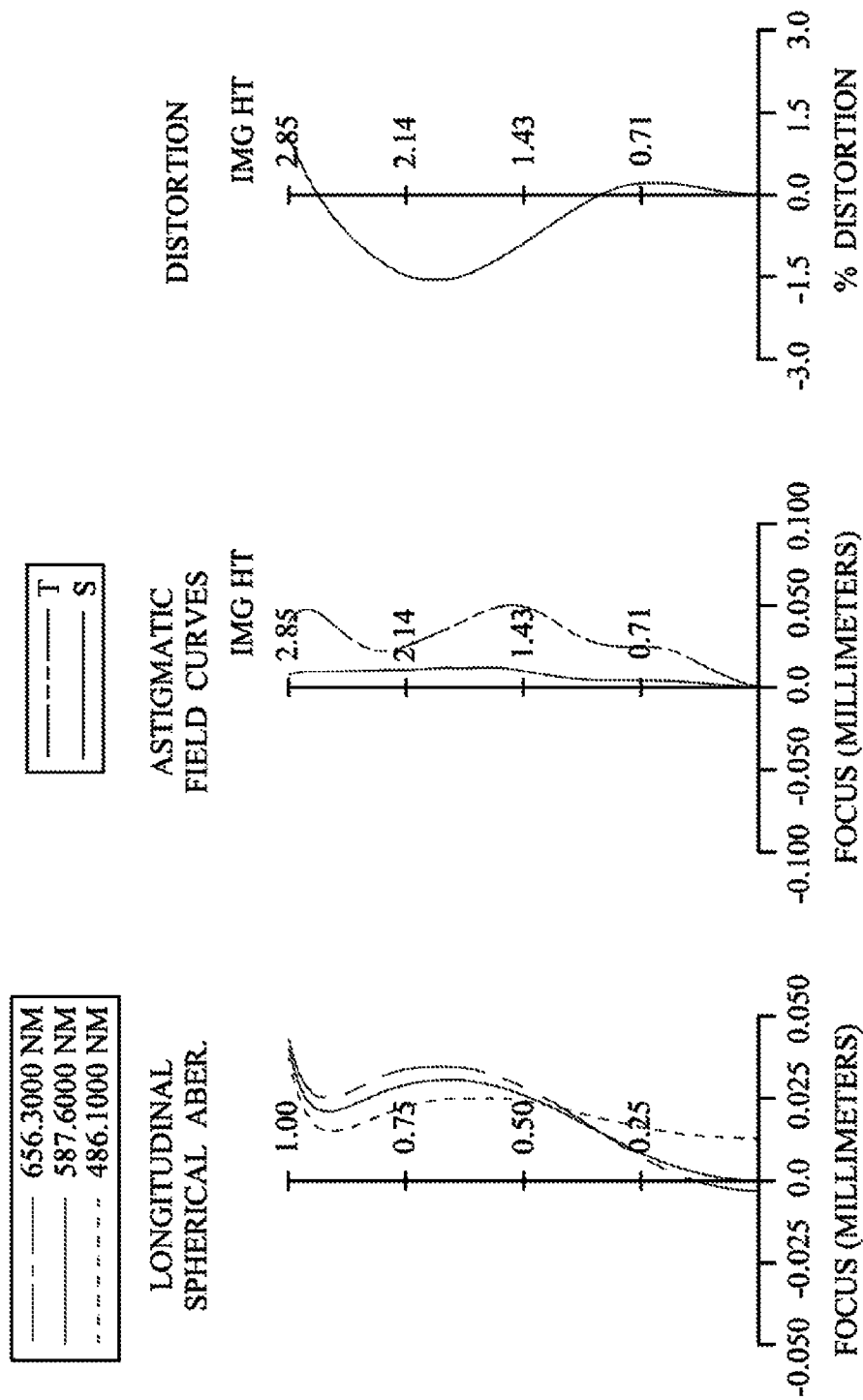


Fig. 14

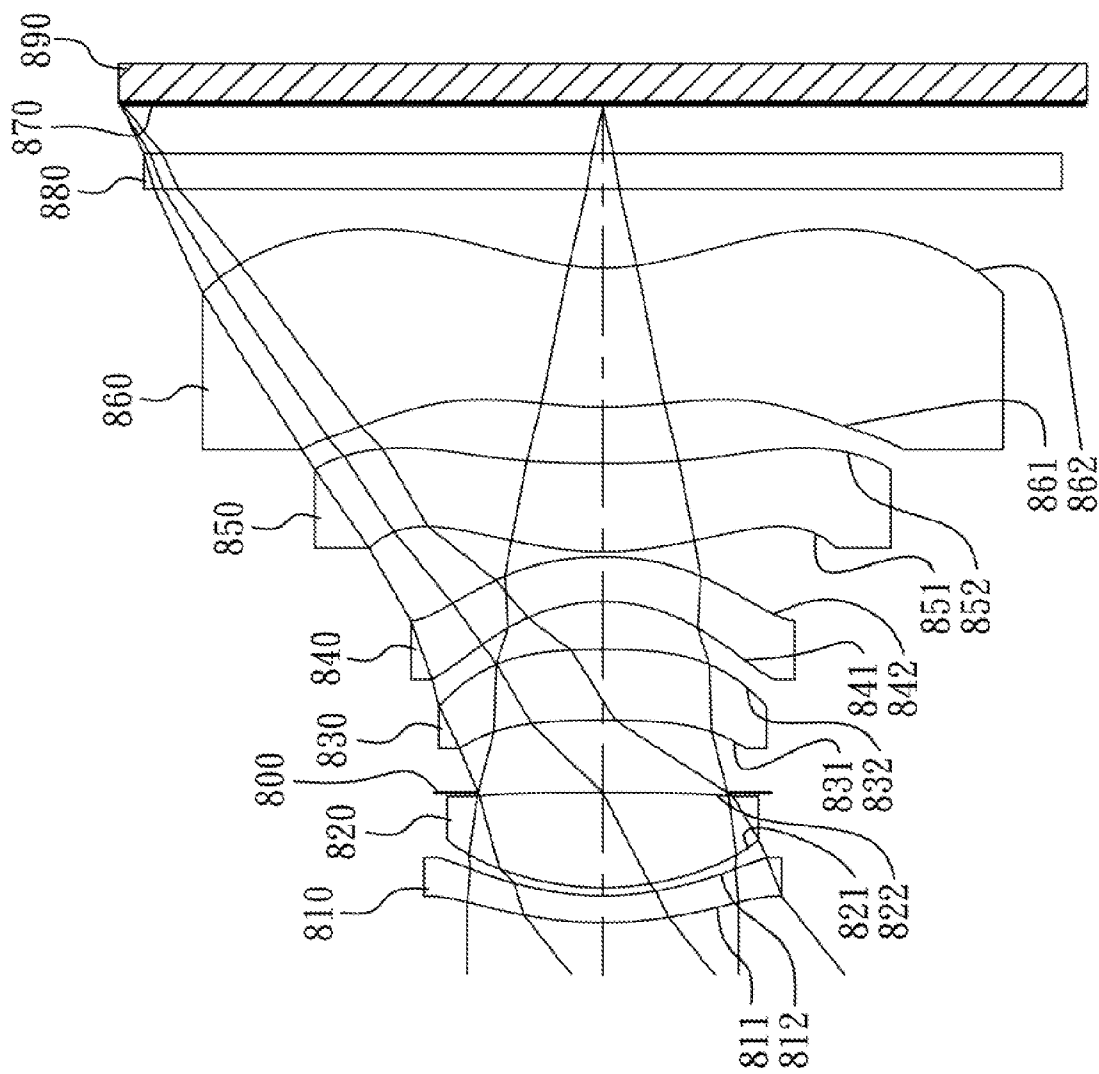


Fig. 15

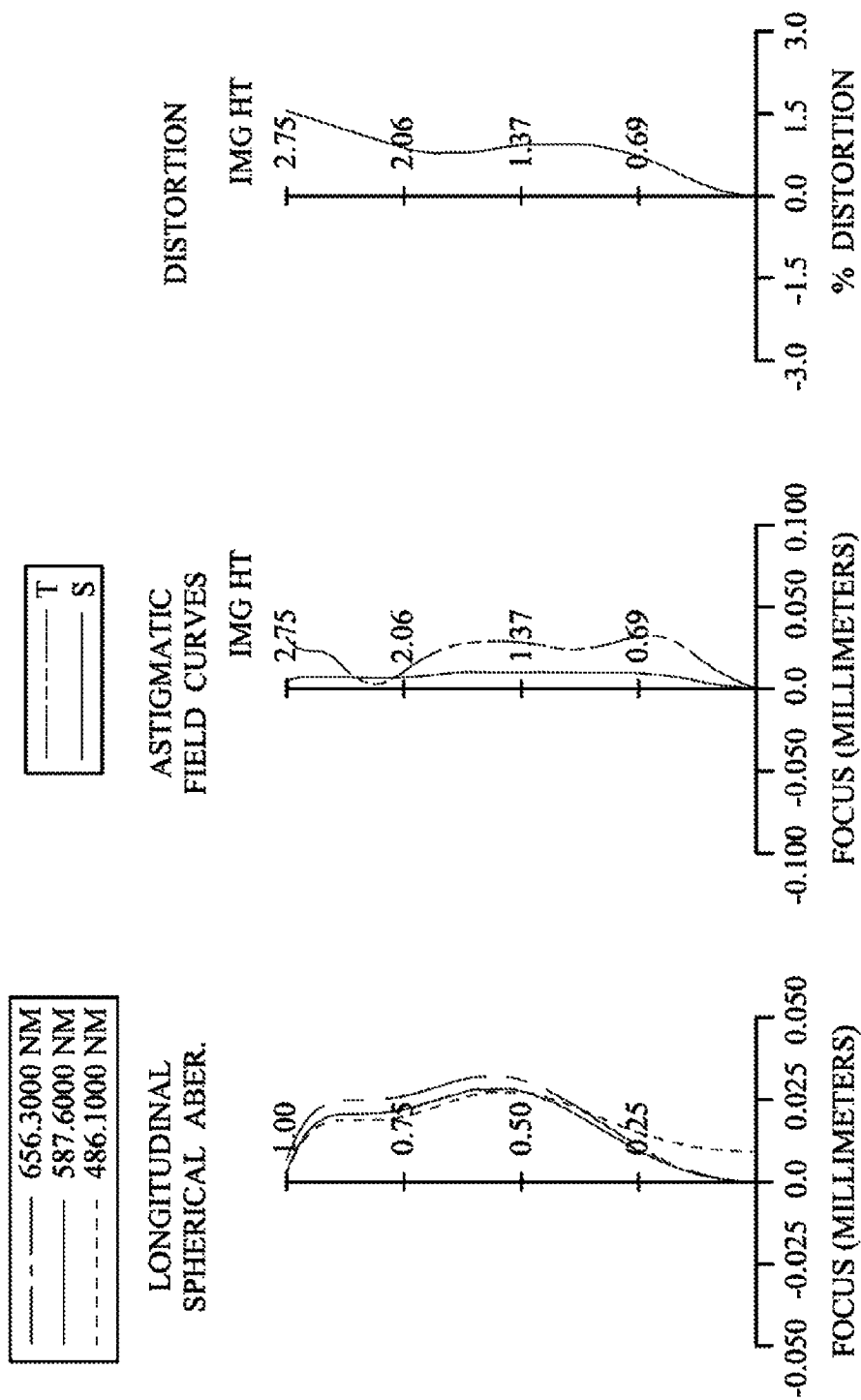


Fig. 16

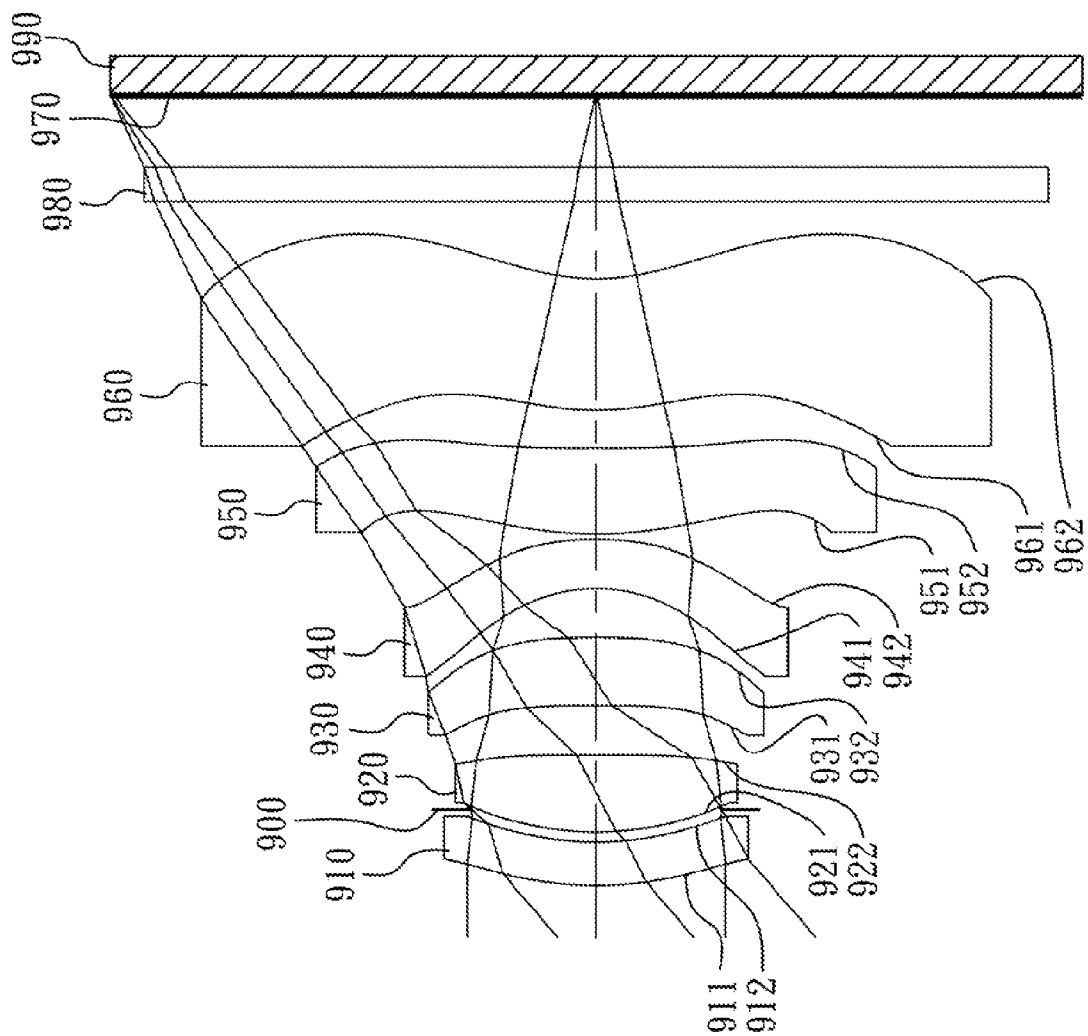
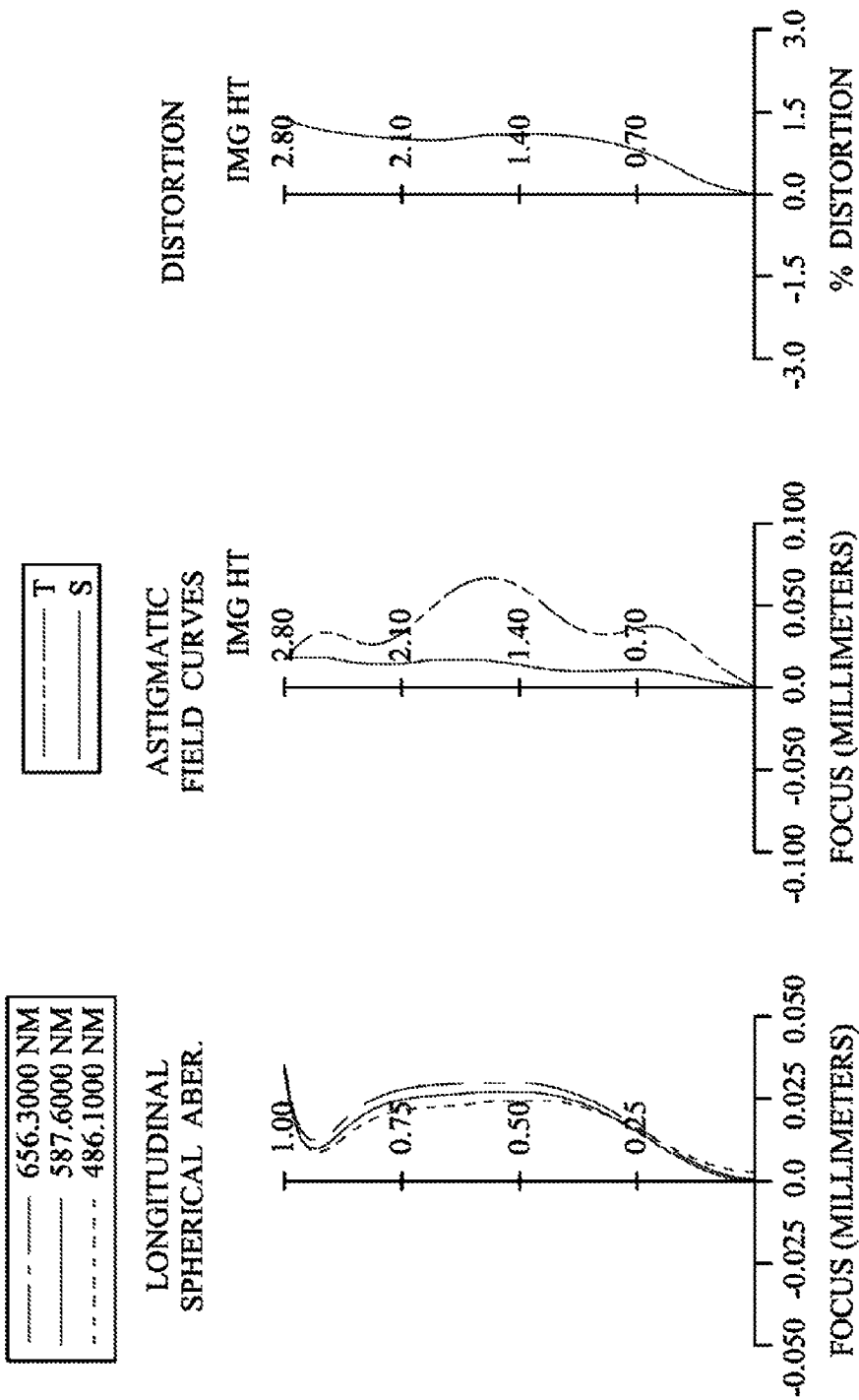
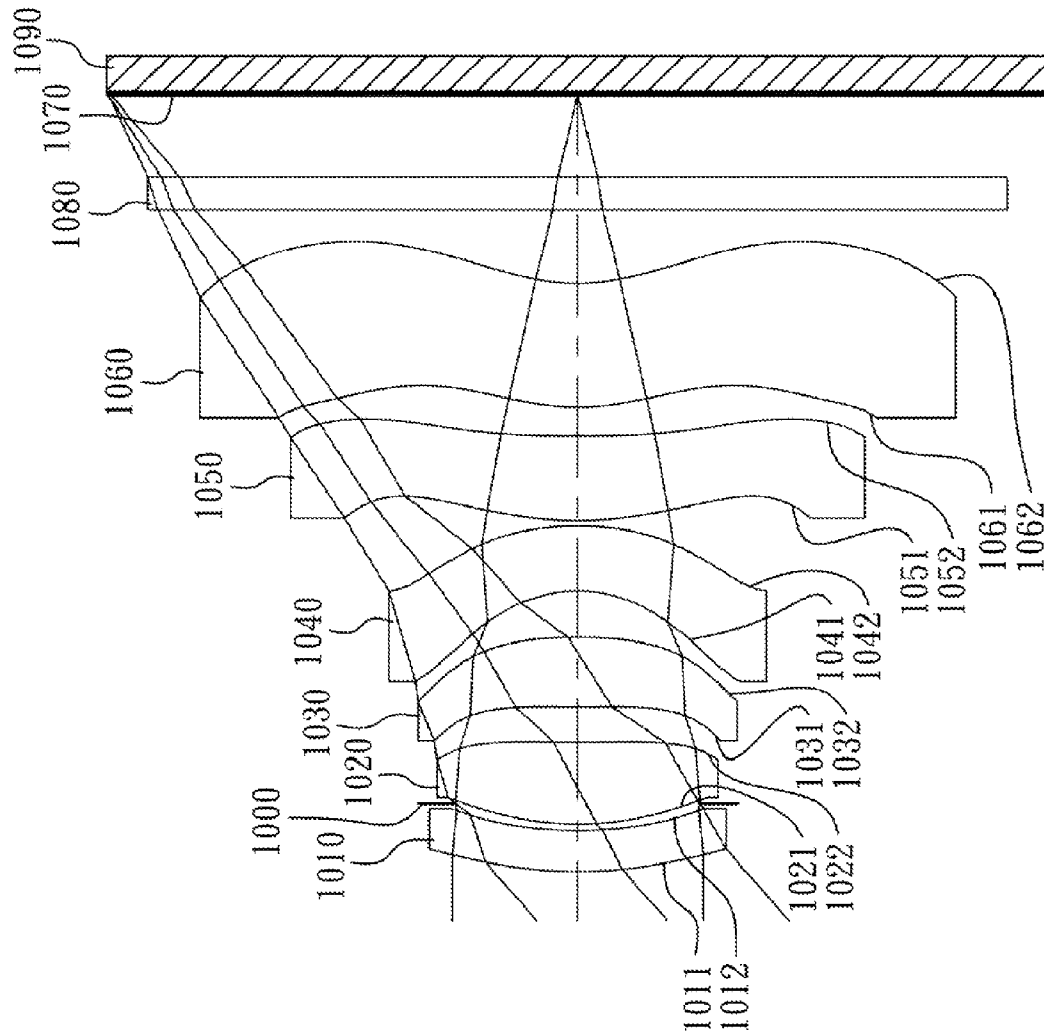


Fig. 17





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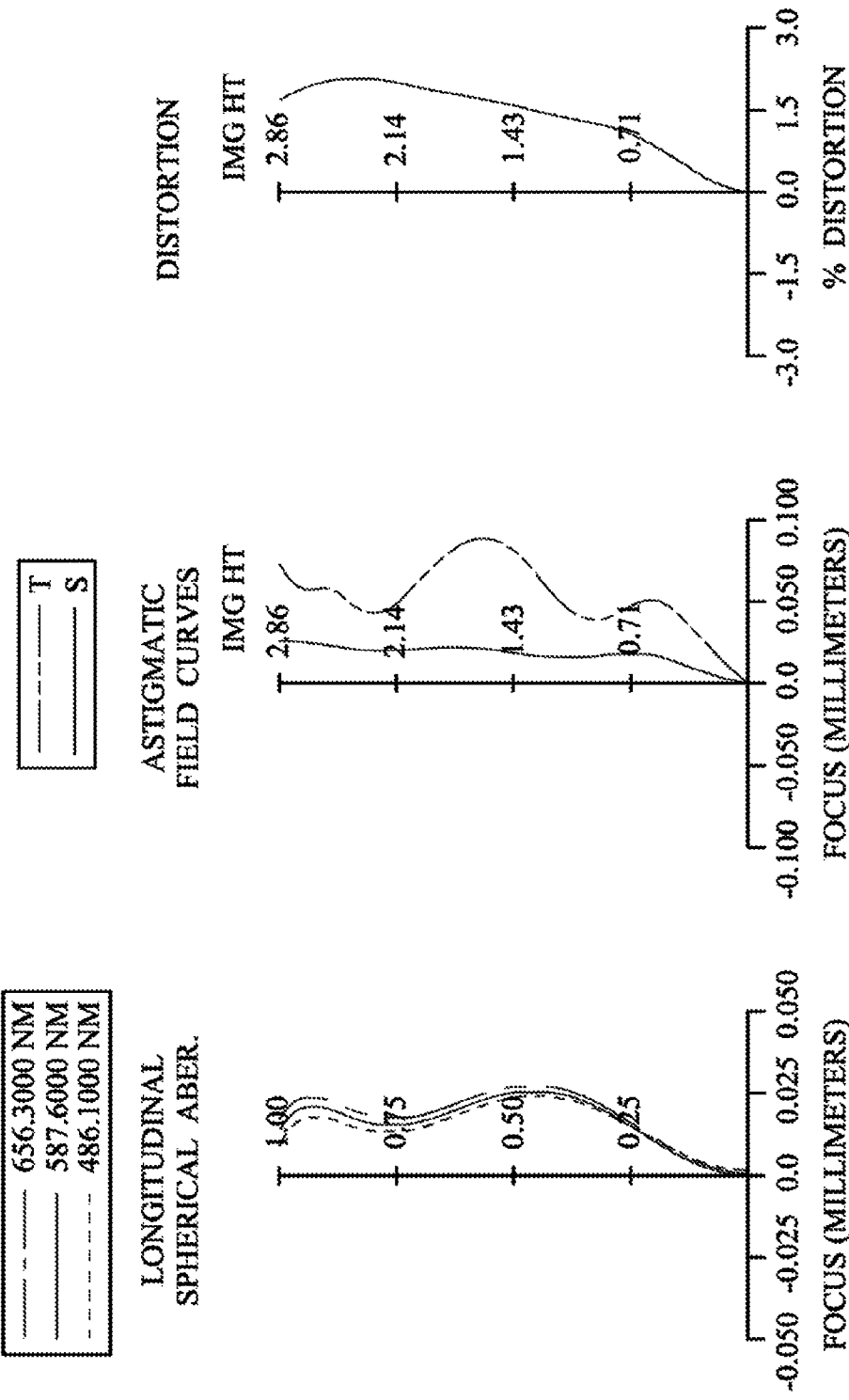


Fig. 20

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IMAGE SYSTEM LENS ASSEMBLY

RELATED APPLICATIONS

The application claims priority to Taiwan Application Serial Number 101124042, filed on Jul. 4, 2012, which is herein incorporated by reference.

BACKGROUND

1. Technical Field

The present invention relates to an image system lens assembly. More particularly, the present invention relates to a compact image system lens assembly applicable to electronic products.

2. Description of Related Art

In recent years, with the popularity of mobile products with camera functionalities, the demand for miniaturizing a photographing lens assembly is increasing. The sensor of a conventional photographing camera is typically a CCD (Charge-Coupled Device) image sensor or a CMOS (Complementary Metal-Oxide Semiconductor) sensor. As advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact photographing lens assemblies have gradually evolved toward the field of higher megapixels, there is an increasing demand for photographing lens assemblies featuring better image quality.

A conventional compact optical lens system employed in a portable electronic product such as disclosed in U.S. Pat. No. 7,869,142, mainly adopts a four-element lens structure. Due to the popularity of mobile products with high-end specifications, such as smart phones and PDAs (Personal Digital Assistants), the higher pixel and image-quality requirements of the compact optical lens system have increased rapidly. However, the conventional four-element lens structure cannot satisfy the requirements of the compact optical lens system.

Although other conventional optical lens system with five-element lens structure such as disclosed in U.S. Pat. No. 8,000,030 and U.S. Pat. No. 8,000,031 which provide better image quality and higher resolution. However, the object side of the optical lens system does not have a configuration of two lens elements that one has negative refractive power and the other has stronger positive refractive power. Thus, the field of view of the optical lens system cannot be enlarged, and the total track length thereof cannot be reduced. Meanwhile, the aberration and the distortion of the optical lens system cannot be eliminated.

SUMMARY

According to one aspect of the present disclosure, an image system lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element with negative refractive power has a convex object-side surface and a concave image-side surface. The second lens element has positive refractive power. The third lens element has refractive power. The fourth lens element with negative refractive power has a concave object-side surface and a convex image-side surface. The fifth lens element with positive refractive power is made of plastic material, wherein at least one of an object-side surface and an image-side surface of the fifth lens element is aspheric. The sixth lens element with refractive power is made of plastic material and has a convex object-side surface and a concave image-side surface, wherein the image-side surface of the sixth lens element changes from concave at

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a paraxial region to convex at a peripheral region, and at least one of the object-side surface and the image-side surface of the sixth lens element is aspheric. When a focal length of the image system lens assembly is f , a focal length of the first lens element is f_1 and a focal length of the second lens element is f_2 , the following relationships are satisfied:

$$-0.8 < f/f_1 < 0; \text{ and}$$

$$0.7 < f/f_2 < 2.4.$$

According to another aspect of the present disclosure, an image system lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with negative refractive power has a convex object-side surface and a concave image-side surface. The second lens element has positive refractive power. The third lens element has refractive power. The fourth lens element has negative refractive power. The fifth lens element with positive refractive power is made of plastic material, wherein at least one of an object-side surface and an image-side surface of the fifth lens element is aspheric. The sixth lens element with refractive power is made of plastic material, and has a concave image-side surface, wherein the image-side surface of the sixth lens element changes from concave at a paraxial region to convex at a peripheral region, and at least one of an object-side surface and the image-side surface of the sixth lens element is aspheric. When a focal length of the image system lens assembly is f , a focal length of the second lens element is f_2 , a sum of central thicknesses of the first through sixth lens elements is ΣCT , and an axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is T_d , the following relationships are satisfied:

$$0.55 < \Sigma CT/T_d < 0.90; \text{ and}$$

$$0.7 < f/f_2 < 2.4.$$

According to yet another aspect of the present disclosure, an image system lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with negative refractive power has a convex object-side surface and a concave image-side surface. The second lens element has positive refractive power. The third lens element has refractive power. The fourth lens element has negative refractive power. The fifth lens element with positive refractive power is made of plastic material, wherein at least one of an object-side surface and an image-side surface of the fifth lens element is aspheric. The sixth lens element with refractive power is made of plastic material, and has a convex object-side surface and a concave image-side surface, wherein the image-side surface of the sixth lens element changes from concave at a paraxial region to convex at a peripheral region, and at least one of the object-side surface and the image-side surface of the sixth lens element is aspheric. When a focal length of the image system lens assembly is f , a focal length of the second lens element is f_2 , and a half of a maximal field of view of the image system lens assembly is HFOV, the following relationships are satisfied:

$$3.0 \text{ mm} < f/\tan(HFOV) < 6.0 \text{ mm}; \text{ and}$$

$$0.7 < f/f_2 < 2.4.$$

The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

FIG. 1 is a schematic view of an image system lens assembly according to the 1st embodiment of the present disclosure;

FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 1st embodiment;

FIG. 3 is a schematic view of an image system lens assembly according to the 2nd embodiment of the present disclosure;

FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 2nd embodiment;

FIG. 5 is a schematic view of an image system lens assembly according to the 3rd embodiment of the present disclosure;

FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 3rd embodiment;

FIG. 7 is a schematic view of an image system lens assembly according to the 4th embodiment of the present disclosure;

FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 4th embodiment;

FIG. 9 is a schematic view of an image system lens assembly according to the 5th embodiment of the present disclosure;

FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 5th embodiment;

FIG. 11 is a schematic view of an image system lens assembly according to the 6th embodiment of the present disclosure;

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 6th embodiment;

FIG. 13 is a schematic view of an image system lens assembly according to the 7th embodiment of the present disclosure;

FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 7th embodiment;

FIG. 15 is a schematic view of an image system lens assembly according to the 8th embodiment of the present disclosure;

FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 8th embodiment;

FIG. 17 is a schematic view of an image system lens assembly according to the 9th embodiment of the present disclosure;

FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 9th embodiment;

FIG. 19 is a schematic view of an image system lens assembly according to the 10th embodiment of the present disclosure; and

FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 10th embodiment.

An image system lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The image system lens assembly further includes an image sensor located on an image plane.

The first lens element with negative refractive power has a convex object-side surface and a concave image-side surface, so that the field of view of the image system lens assembly can be enlarged by properly adjusting the negative refractive power of the first lens element, and the curvatures of the object-side surface and the image-side surface of the first lens element.

The second lens element has positive refractive power which is stronger than the refractive power of the first lens element. Therefore, the excessive back focal length of the image system lens assembly which is caused by the negative refractive power of the first lens element can be reduced by stronger positive refractive power of the second lens element. The second lens element can have a convex object-side surface, therefore, the aberration and the distortion of the image system lens assembly can be reduced.

The third lens element has refractive power. When the third lens element has positive refractive power, the sensitivity of the image system lens assembly can be reduced. When the third lens element has negative refractive power, the aberration of the image system lens assembly can be corrected.

The fourth lens element with negative refractive power has a concave object-side surface and a convex image-side surface. Therefore, the aberration and the astigmatism of the image system lens assembly can be corrected.

The fifth lens element with positive refractive power is made of plastic material, and can have a convex object-side surface and a concave image-side surface. Therefore, the high order aberration of the image system lens assembly can be corrected by properly adjusting the refractive power of the fifth lens element for improving the resolution so as to obtain better image quality. The object-side surface of the fifth lens element changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface of the fifth lens element changes from concave at a paraxial region to convex at a peripheral region. Therefore, the angle at which the incident light projects onto the image sensor from the off-axis field can be effectively reduced and the aberration can be corrected as well.

The sixth lens element with refractive power is made of plastic material and has a convex object-side surface and a concave image-side surface, wherein the image-side surface of the sixth lens element changes from concave at a paraxial region to convex at a peripheral region. Therefore, a principal point of the image system lens assembly can be positioned away from the image plane, and the back focal length of the image system lens assembly can be reduced so as to maintain the compact size thereof. Moreover, the incident angle of the off-axis field on the image sensor can be effectively minimized and the aberration can be corrected as well.

When a focal length of the image system lens assembly is f , and a focal length of the first lens element is f_1 , the following relationship is satisfied: $-0.8 < f/f_1 < 0$. Therefore, the field of view of the image system lens assembly can be enlarged by properly adjusting the negative refractive power of the first lens element.

When the focal length of the image system lens assembly is f , and a focal length of the second lens element is f_2 , the following relationship is satisfied: $0.7 < f/f_2 < 2.4$. Therefore,

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the excessive back focal length of the image system lens assembly which is caused by the negative refractive power of the first lens element can be reduced by larger refractive power of the second lens element. Preferably, f and f_2 can satisfy the following relationship: $1.0 < f/f_2 < 1.8$.

When the focal length of the image system lens assembly is f , the focal length of the first lens element is f_1 , the focal length of the second lens element is f_2 , a focal length of the third lens element is f_3 , a focal length of the fourth lens element is f_4 , a focal length of the fifth lens element is f_5 , and a focal length of the sixth lens element is f_6 , the following relationship is satisfied:

$$f/f_2 > f/f_i, i=1,3,4,5,6,$$

Therefore, the refractive power of the second lens element is strongest, thus the back focal length of the image system lens assembly can be reduced for maintaining the compact size thereof.

When an axial distance between the first lens element and the second lens element is T_{12} , an axial distance between the second lens element and the third lens element is T_{23} , an axial distance between the third lens element and the fourth lens element is T_{34} , an axial distance between the fourth lens element and the fifth lens element is T_{45} , and an axial distance between the fifth lens element and the sixth lens element is T_{56} , the following relationships are satisfied: $T_{23} > T_{12}$; $T_{23} > T_{34}$; $T_{23} > T_{45}$ and $T_{23} > T_{56}$. Therefore, the assembling of the image system lens assembly is easier due to the proper adjustment of the distance between the lens elements, and the manufacturing yield rate of the image system lens assembly can be improved.

When an Abbe number of the fourth lens element is V_4 , and an Abbe number of the fifth lens element is V_5 , the following relationship is satisfied: $0.20 < V_4/V_5 < 0.60$. Therefore, the chromatic aberration of the image system lens assembly can be reduced.

When the focal length of the image system lens assembly is f , and a half of a maximal field of view of the image system lens assembly is $HFOV$, the following relationship is satisfied: $3.0 \text{ mm} < f/\tan(HFOV) < 6.0 \text{ mm}$. Therefore, the larger field of view of the image system lens assembly can be obtained.

When a curvature radius of the object-side surface of the sixth lens element is R_{11} , and a curvature radius of the image-side surface of the sixth lens element is R_{12} , the following relationship is satisfied: $-0.10 < (R_{11}-R_{12})/(R_{11}+R_{12}) < 0.45$. Therefore, the astigmatism can be corrected by properly adjusting the curvature of the surfaces of the sixth lens element.

When a sum of central thicknesses of the first through sixth lens elements is ΣCT , and an axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is T_d , the following relationship is satisfied: $0.55 < \Sigma CT/T_d < 0.90$. Therefore, it is favorable for forming and assembling the lens elements of the image system lens assembly by adjusting the thickness of the lens elements, and the fabrication yield rate can be increased. Moreover, the total track length of the image system lens assembly can be reduced for maintaining the compact size thereof. Preferably, ΣCT and T_d can satisfy the following relationship: $0.65 < \Sigma CT/T_d < 0.90$.

When a maximum image height of the image system lens assembly is $ImgH$, and an axial distance between the object-side surface of the first lens element and the image plane is TTL , the following relationship is satisfied: $TTL/ImgH < 1.8$. Therefore, the compact size of the image system lens assembly can be maintained.

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According to the image system lens assembly of the present disclosure, the lens elements thereof can be made of plastic or glass materials. When the lens elements are made of glass material, the allocation of the refractive power of the image system lens assembly may be more flexible and easier to design. When the lens elements are made of plastic material, the manufacturing costs can be effectively reduced. Furthermore, the surface of each lens element can be aspheric, so that it is easier to make the surface into non-spherical shapes. As a result, more controllable variables are obtained for reducing aberrations, and the number of required lens elements for constructing an image system lens assembly can be reduced. Therefore, the total track length of the image system lens assembly can also be reduced.

According to the image system lens assembly of the present disclosure, each of an object-side surface and an image-side surface of every lens element has a paraxial region and a peripheral region. The paraxial region refers to the region of the surface where light rays travel close to an optical axis and the peripheral region refers to the region of the surface where light rays travel away from the optical axis. Particularly, when a lens element has a convex surface, it indicates that the surface is convex at the paraxial region; and when the lens element has a concave surface, it indicates that the surface is concave at the paraxial region.

According to the image system lens assembly of the present disclosure, there can be at least one stop provided, such as an aperture stop, a glare stop, or a field stop. Said glare stop or said field stop is allocated for reducing the stray light and thereby improving the image resolution thereof. Furthermore, an aperture stop can be configured as a front stop or a middle stop. A front stop can provide a longer distance from an exit pupil of the image system lens assembly to an image plane and thereby the generated telecentric effect improves the image-sensing efficiency of an image sensor. A middle stop is favorable for enlarging the field of view of the image system lens assembly and thereby provides a wider field of view for the same.

The image system lens assembly of the present disclosure has advantages on superb aberration correction and excellent image quality. Therefore, the image system lens assembly can be applicable on the electronic image systems such as 3D (three dimensional) capturing systems, digital cameras, portable devices, and tablet computers, etc.

According to the above description of the present disclosure, the following 1st-10th specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1 is a schematic view of an image system lens assembly according to the 1st embodiment of the present disclosure. FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 1st embodiment. In FIG. 1, the image system lens assembly includes, in order from an object side to an image side, a first lens element 110, an aperture stop 100, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 180, an image plane 170, and an image sensor 190.

The first lens element 110 with negative refractive power has a convex object-side surface 111 and a concave image-side surface 112. The first lens element 110 is made of plastic material and has the object-side surface 111 and the image-side surface 112 being aspheric.

The second lens element **120** with positive refractive power has a convex object-side surface **121** and a concave image-side surface **122**. The second lens element **120** is made of plastic material and has the object-side surface **121** and the image-side surface **122** being aspheric.

The third lens element **130** with positive refractive power has a concave object-side surface **131** and a convex image-side surface **132**. The third lens element **130** is made of plastic material and has the object-side surface **131** and the image-side surface **132** being aspheric.

The fourth lens element **140** with negative refractive power has a concave object-side surface **141** and a convex image-side surface **142**. The fourth lens element **140** is made of plastic material and has the object-side surface **141** and the image-side surface **142** being aspheric.

The fifth lens element **150** with positive refractive power has a convex object-side surface **151** and a concave image-side surface **152**, wherein the object-side surface **151** of the fifth lens element **150** changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface **152** of the fifth lens element **150** changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element **150** is made of plastic material and has the object-side surface **151** and the image-side surface **152** being aspheric.

The sixth lens element **160** with positive refractive power has a convex object-side surface **161** and a concave image-side surface **162**, wherein the image-side surface **162** of the sixth lens element **160** changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element **160** is made of plastic material and has the object-side surface **161** and the image-side surface **162** being aspheric.

The IR-cut filter **180** is made of glass material, wherein the IR-cut filter **180** is located between the sixth lens element **160** and the image plane **170**, and will not affect the focal length of the image system lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (A_i) \times (Y^i)$$

where:

X is the relative distance of a point on the aspheric surface spaced at a distance Y from the optical axis relative to the tangential plane at the aspheric surface vertex on the optical axis;

Y is the distance from the point on the curve of the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

A_i is the i-th aspheric coefficient.

In the image system lens assembly according to the 1st embodiment, when a focal length of the image system lens assembly is f, an f-number of the image system lens assembly is Fno, and a half of a maximal field of view of the image system lens assembly is HFOV, these parameters have the following values:

f=3.48 mm;

Fno=2.35; and

HFOV=39.0 degrees.

In the image system lens assembly according to the 1st embodiment, when an Abbe number of the fourth lens element **140** is V4, and an Abbe number of the fifth lens element **150** is V5, the following relationship is satisfied:

$$V4/V5=0.43.$$

In the image system lens assembly according to the 1st embodiment, when a curvature radius of the object-side surface **161** of the sixth lens element **160** is R11, and a curvature radius of the image-side surface **162** of the third lens element **160** is R12, the following relationship is satisfied:

$$(R11-R12)/(R11+R12)=0.08.$$

In the image system lens assembly according to the 1st embodiment, when the focal length of the image system lens assembly is f, and a focal length of the first lens element **110** is f1, the following relationship is satisfied: f/f1=-0.43.

In the image system lens assembly according to the 1st embodiment, when the focal length of the image system lens assembly is f, and a focal length of the second lens element **120** is f2, the following relationship is satisfied: f/f2=1.41.

In the image system lens assembly according to the 1st embodiment, a sum of central thicknesses of the first through sixth lens elements (**110**, **120**, **130**, **140**, **150**, **160**) is ΣCT, and an axial distance between the object-side surface **111** of the first lens element **110** and the image-side surface **162** of the sixth lens element **160** is Td, the following relationship is satisfied: ΣCT/Td=0.76.

In the image system lens assembly according to the 1st embodiment, when the focal length of the image system lens assembly is f, and the half of the maximal field of view of the image system lens assembly is HFOV, the following relationship is satisfied: f/tan(HFOV)=4.29 mm.

In the image system lens assembly according to the 1st embodiment, wherein a maximum image height of the image system lens assembly is ImgH which here is a half of the diagonal length of the photosensitive area of the image sensor **190** on the image plane **170**, and an axial distance between the object-side surface **111** of the first lens element **110** and an image plane **170** is TTL, the following relationship is satisfied: TTL/ImgH=1.66.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment f = 3.48 mm, Fno = 2.35, HFOV = 39.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Lens 1	2.040 (ASP)	0.205	Plastic	1.634	23.8	-8.14	
2		1.405 (ASP)	0.204					
3	Ape. Stop	Plano	-0.143					
4	Lens 2	1.338 (ASP)	0.452	Plastic	1.544	55.9	2.46	

TABLE 1-continued

1st Embodiment f = 3.48 mm, Fno = 2.35, HFOV = 39.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
5		2824.909 (ASP)	0.320					
6	Lens 3	-8.719 (ASP)	0.401	Plastic	1.544	55.9	17.79	
7		-4.662 (ASP)	0.235					
8	Lens 4	-0.884 (ASP)	0.250	Plastic	1.634	23.8	-3.05	
9		-1.808 (ASP)	0.030					
10	Lens 5	1.667 (ASP)	0.350	Plastic	1.544	55.9	4.54	
11		4.749 (ASP)	0.222					
12	Lens 6	2.043 (ASP)	1.148	Plastic	1.544	55.9	62.21	
13		1.743 (ASP)	0.492					
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—	
15		Plano	0.435					
16	Image	Plano	—					

Reference wavelength (d-line) is 587.6 nm

TABLE 2

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	1.6472E-01	-7.5791E+00	-4.8717E+00	0.0000E+00	0.0000E+00	1.3127E+00
A4 =	-1.2919E-01	-1.3043E-01	-1.6992E-01	-1.5859E-01	-2.6870E-01	-2.7551E-01
A6 =	3.0549E-01	8.9705E-01	7.9677E-01	-7.0775E-02	-3.7395E-02	-7.6013E-02
A8 =	-5.1378E-01	-1.8374E+00	-1.5780E+00	2.3790E-01	-3.3031E-02	-1.8623E-02
A10 =	3.7786E-01	1.8588E+00	1.5716E+00	-2.5295E-01	2.1918E-01	2.6555E-01
A12 =	-1.5706E-01	-7.8278E-01	-4.4901E-01	-1.0314E-01	2.4226E-01	-9.4393E-02
A14 =				6.5730E-01	-3.0268E-01	-4.5524E-02

Surface #						
	8	9	10	11	12	13
k =	-2.9075E+00	1.6868E-01	-9.2581E+00	3.0000E+00	-1.5378E+01	-2.3992E+00
A4 =	-3.5110E-01	-3.0448E-01	7.8826E-02	1.7557E-01	-1.9002E-01	-1.8222E-01
A6 =	1.0562E-01	3.3875E-01	-1.9025E-01	-3.8131E-01	3.5324E-02	9.7623E-02
A8 =	5.0913E-01	1.9305E-02	9.2275E-02	3.0878E-01	1.8523E-02	-4.2574E-02
A10 =	-6.7679E-01	-1.2167E-01	-3.0406E-02	-1.7157E-01	-1.0336E-02	1.2934E-02
A12 =	5.3047E-01	4.7918E-02	-9.2850E-04	6.0698E-02	2.2036E-03	-2.4524E-03
A14 =	-1.8280E-01	1.3705E-02	2.3757E-03	-1.1501E-02	-2.3069E-04	2.5592E-04
A16 =				8.6055E-04	9.7129E-06	-1.1161E-05

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A1-A16 represent the aspheric coefficients ranging from the 1st order to the 16th order. This information related to Table 1 and Table 2 applies also to the Tables for the remaining embodiments, and so an explanation in this regard will not be provided again.

2nd Embodiment

FIG. 3 is a schematic view of an image system lens assembly according to the 2nd embodiment of the present disclosure. FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 2nd embodiment. In FIG. 3, the image system lens assembly includes, in order from an object side to an image side, a first lens element 210, an aperture stop 200, a second lens element 220, a third lens element 230, a

fourth lens element 240, a fifth lens element 250, a sixth lens element 260, an IR-cut filter 280, an image plane 270 and an image sensor 290.

The first lens element 210 with negative refractive power has a convex object-side surface 211 and a concave image-side surface 212. The first lens element 210 is made of plastic material and has the object-side surface 211 and the image-side surface 212 being aspheric.

The second lens element 220 with positive refractive power has a convex object-side surface 221 and a convex image-side surface 222. The second lens element 220 is made of plastic material and has the object-side surface 221 and the image-side surface 222 being aspheric.

The third lens element 230 with positive refractive power has a concave object-side surface 231 and a convex image-side surface 232. The third lens element 230 is made of plastic material and has the object-side surface 231 and the image-side surface 232 being aspheric.

The fourth lens element 240 with negative refractive power has a concave object-side surface 241 and a convex image-side surface 242. The fourth lens element 240 is made of

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plastic material and has the object-side surface **241** and the image-side surface **242** being aspheric.

The fifth lens element **250** with positive refractive power has a convex object-side surface **251** and a concave image-side surface **252**, wherein the object-side surface **251** of the fifth lens element **250** changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface **252** of the fifth lens element **250** changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element **260** is made of plastic material and has the object-side surface **251** and the image-side surface **252** being aspheric.

The sixth lens element **260** with negative refractive power has a convex object-side surface **261** and a concave image-

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side surface **262**, wherein the image-side surface **262** of the sixth lens element **260** changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element **260** is made of plastic material and has the object-side surface **261** and the image-side surface **262** being aspheric.

The IR-cut filter **280** is made of glass material, wherein the IR-cut filter **280** is located between the sixth lens element **260** and the image plane **270**, and will not affect the focal length of the image system lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment								
f = 3.58 mm, Fno = 2.25, HFOV = 38.2 deg.								
Surface #		Curvature Radius		Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano		Infinity				
1	Lens 1	2.010	(ASP)	0.186	Plastic	1.640	23.3	−7.88
2		1.385	(ASP)	0.225				
3	Ape. Stop	Plano		−0.176				
4	Lens 2	1.362	(ASP)	0.484	Plastic	1.544	55.9	2.40
5		−29.180	(ASP)	0.386				
6	Lens 3	−6.974	(ASP)	0.319	Plastic	1.544	55.9	169.17
7		−6.587	(ASP)	0.207				
8	Lens 4	−1.056	(ASP)	0.250	Plastic	1.640	23.3	−3.76
9		−2.056	(ASP)	0.030				
10	Lens 5	1.619	(ASP)	0.330	Plastic	1.544	55.9	4.45
11		4.536	(ASP)	0.328				
12	Lens 6	2.204	(ASP)	1.064	Plastic	1.535	56.3	−30.42
13		1.615	(ASP)	0.450				
14	IR-cut filter	Plano		0.200	Glass	1.517	64.2	—
15		Plano		0.480				
16	Image	Plano		—				

Reference wavelength (d-line) is 587.6 nm

TABLE 4

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	-2.4152E-01	-7.3574E+00	-4.8121E+00	0.0000E+00	0.0000E+00	3.0000E+00
A4 =	-1.3645E-01	-1.3058E-01	-1.6634E-01	-1.1428E-01	-2.8100E-01	-3.1542E-01
A6 =	3.0178E-01	8.9077E-01	8.4419E-01	-7.6876E-02	-2.3605E-03	-4.7101E-02
A8 =	-5.1267E-01	-1.8635E+00	-1.6205E+00	2.3489E-01	-3.8961E-02	-7.3475E-03
A10 =	3.5517E-01	1.7971E+00	1.4741E+00	-2.1480E-01	1.7937E-01	2.6066E-01
A12 =	-1.2227E-01	-7.1153E-01	-4.0218E-01	-1.3542E-01	1.9820E-01	-1.0989E-01
A14 =				4.7318E-01	-2.4117E-01	-3.3130E-02
Surface #						
	8	9	10	11	12	13
k =	-3.7723E+00	2.5026E-01	-8.6057E+00	3.0000E+00	-1.4098E+01	-2.6292E+00
A4 =	-3.3722E-01	-2.9325E-01	7.8826E-02	1.7557E-01	-1.9002E-01	-1.8222E-01
A6 =	9.7329E-02	3.1764E-01	-1.9025E-01	-3.8131E-01	3.5324E-02	9.7623E-02
A8 =	4.8557E-01	8.5644E-03	9.2275E-02	3.0878E-01	1.8523E-02	-4.2574E-02
A10 =	-6.8448E-01	-1.1897E-01	-3.0406E-02	-1.7157E-01	-1.0336E-02	1.2934E-02
A12 =	5.4015E-01	5.1469E-02	-9.2850E-04	6.0698E-02	2.2036E-03	-2.4524E-03
A14 =	-1.7634E-01	1.2937E-02	2.3757E-03	-1.1501E-02	-2.3069E-04	2.5592E-04
A16 =				8.6055E-04	9.7129E-06	-1.1161E-05

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In the image system lens assembly according to the 2nd embodiment, the definitions of f , F_{no} , $HFOV$, $V4$, $V5$, $R11$, $R12$, $f1$, $f2$, ΣCT , Td , TL , and $ImgH$ are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment. Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following relationships:

f (mm)	3.58	$f/f1$	-0.45
F_{no}	2.25	$f/f2$	1.49
$HFOV$ (deg.)	38.2	$\Sigma CT/Td$	0.72
$V4/V5$	0.42	$f/\tan(HFOV)$ (mm)	4.54
$(R11 - R12)/(R11 + R12)$	0.15	$TTL/ImgH$	1.64

3rd Embodiment

FIG. 5 is a schematic view of an image system lens assembly according to the 3rd embodiment of the present disclosure. FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 3rd embodiment. In FIG. 5, the image system lens assembly includes, in order from an object side to an image side, a first lens element 310, an aperture stop 300, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, an IR-cut filter 380, an image plane 370, and an image sensor 390.

The first lens element 310 with negative refractive power has a convex object-side surface 311 and a concave image-side surface 312. The first lens element 310 is made of plastic material and has the object-side surface 311 and the image-side surface 312 being aspheric.

The second lens element 320 with positive refractive power has a convex object-side surface 321 and a convex image-side

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surface 322. The second lens element 320 is made of plastic material and has the object-side surface 321 and the image-side surface 322 being aspheric.

The third lens element 330 with negative refractive power has a concave object-side surface 331 and a convex image-side surface 332. The third lens element 330 is made of plastic material and has the object-side surface 331 and the image-side surface 332 being aspheric.

The fourth lens element 340 with negative refractive power has a concave object-side surface 341 and a convex image-side surface 342. The fourth lens element 340 is made of plastic material and has the object-side surface 341 and the image-side surface 342 being aspheric.

The fifth lens element 350 with positive refractive power has a convex object-side surface 351 and a concave image-side surface 352, wherein the object-side surface 351 of the fifth lens element 350 changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface 352 of the fifth lens element 350 changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element 350 is made of plastic material and has the object-side surface 351 and the image-side surface 352 being aspheric.

The sixth lens element 360 with negative refractive power has a convex object-side surface 361 and a concave image-side surface 362, wherein the image-side surface 362 of the sixth lens element 360 changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element 360 is made of plastic material and has the object-side surface 361 and the image-side surface 362 being aspheric.

The IR-cut filter 380 is made of glass material, wherein the IR-cut filter 380 is located between the sixth lens element 360 and the image plane 370, and will not affect the focal length of the image system lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment							
$f = 3.52$ mm, $F_{no} = 2.34$, $HFOV = 38.4$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	1.623 (ASP)	0.177	Plastic	1.607	26.6	-6.74
2		1.114 (ASP)	0.212				
3	Ape. Stop	Plano	-0.171				
4	Lens 2	1.326 (ASP)	0.447	Plastic	1.544	55.9	2.27
5		-16.030 (ASP)	0.463				
6	Lens 3	-6.164 (ASP)	0.241	Plastic	1.535	56.3	-51.40
7		-8.055 (ASP)	0.178				
8	Lens 4	-1.163 (ASP)	0.252	Plastic	1.640	23.3	-3.82
9		-2.409 (ASP)	0.030				
10	Lens 5	1.417 (ASP)	0.330	Plastic	1.535	56.3	3.83
11		4.235 (ASP)	0.423				
12	Lens 6	2.036 (ASP)	0.878	Plastic	1.535	56.3	-21.39
13		1.468 (ASP)	0.450				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.490				
16	Image	Plano	—				

Reference wavelength (d-line) is 587.6 nm

TABLE 6

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	-1.4163E+00	-4.5009E+00	-2.7391E+00	0.0000E+00	0.0000E+00	3.0000E+00
A4 =	-1.6140E-01	-1.2151E-01	-1.9312E-01	-8.7566E-02	-3.1547E-01	-3.6196E-01
A6 =	3.1903E-01	8.2690E-01	8.1275E-01	-8.8219E-02	6.1631E-03	-6.6279E-02
A8 =	-6.3013E-01	-1.9478E+00	1.4678E+00	3.6298E-01	-6.4762E-02	-3.2802E-02
A10 =	4.5206E-01	1.8907E+00	1.2654E+00	-2.6491E-01	2.1395E-01	2.8514E-01
A12 =	-1.5453E-01	-8.2872E-01	-1.9403E-01	-3.5171E-01	2.9963E-01	-6.3534E-02
A14 =				1.1582E+00	-2.8234E-01	-6.0641E-02

Surface #						
	8	9	10	11	12	13
k =	-5.3515E+00	4.9833E-01	-8.6699E+00	3.0000E+00	-1.0270E+01	-2.9297E+00
A4 =	-3.2128E-01	-3.1742E-01	7.8826E-02	1.7557E-01	-1.9002E-01	-1.8222E-01
A6 =	4.8207E-02	3.4449E-01	-1.9025E-01	-3.8131E-01	3.5324E-02	9.7623E-02
A8 =	4.8180E-01	-2.5932E-02	9.2275E-02	3.0878E-01	1.8523E-02	-4.2574E-02
A10 =	-6.8166E-01	-1.2120E-01	-3.0406E-02	-1.7157E-01	-1.0336E-02	1.2934E-02
A12 =	5.4461E-01	6.8924E-02	-9.2850E-04	6.0698E-02	2.2036E-03	-2.4524E-03
A14 =	-1.7112E-01	7.5956E-03	2.3757E-03	-1.1501E-02	-2.3069E-04	2.5592E-04
A16 =				8.6055E-04	9.7129E-06	-1.1161E-05

In the image system lens assembly according to the 3rd embodiment, the definitions of f, Fno, HFOV, V4, V5, R11, R12, f1, f2, Σ CT, Td, TTL, and ImgH are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment. Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following relationships:

f (mm)	3.52	f/f1	-0.52
Fno	2.34	f/f2	1.55
HFOV (deg.)	38.4	Σ CT/Td	0.67
V4/V5	0.41	f/tan(HFOV) (mm)	4.45
(R11 - R12)/(R11 + R12)	0.16	TTL/ImgH	1.59

4th Embodiment

FIG. 7 is a schematic view of an image system lens assembly according to the 4th embodiment of the present disclosure. FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 4th embodiment. In FIG. 7, the image system lens assembly includes, in order from an object side to an image side, a first lens element 410, an aperture stop 400, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, an IR-cut filter 480, an image plane 470, and an image sensor 490.

The first lens element 410 with negative refractive power has a convex object-side surface 411 and a concave image-side surface 412. The first lens element 410 is made of plastic material and has the object-side surface 411 and the image-side surface 412 being aspheric.

The second lens element 420 with positive refractive power has a convex object-side surface 421 and a concave image-side surface 422. The second lens element 420 is made of

plastic material and has the object-side surface 421 and the image-side surface 422 being aspheric.

The third lens element 430 with positive refractive power has a concave object-side surface 431 and a convex image-side surface 432. The third lens element 430 is made of plastic material and has the object-side surface 431 and the image-side surface 432 being aspheric.

The fourth lens element 440 with negative refractive power has a concave object-side surface 441 and a convex image-side surface 442. The fourth lens element 440 is made of plastic material and has the object-side surface 441 and the image-side surface 442 being aspheric.

The fifth lens element 450 with positive refractive power has a convex object-side surface 451 and a concave image-side surface 452, wherein the object-side surface 451 of the fifth lens element 450 changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface 452 of the fifth lens element 450 changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element 450 is made of plastic material and has the object-side surface 451 and the image-side surface 452 being aspheric.

The sixth lens element 460 with negative refractive power has a convex object-side surface 461 and a concave image-side surface 462, wherein the image-side surface 462 of the sixth lens element 460 changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element 460 is made of plastic material and has the object-side surface 461 and the image-side surface 462 being aspheric.

The IR-cut filter 480 is made of glass material, wherein the IR-cut filter 480 is located between the sixth lens element 460 and the image plane 470, and will not affect the focal length of the image system lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment f = 3.42 mm, Fno = 2.35, HFOV = 39.4 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	2.095 (ASP)	0.227	Plastic	1.633	23.4	-8.67
2		1.453 (ASP)	0.195				
3	Ape. Stop	Plano	-0.131				
4	Lens 2	1.296 (ASP)	0.441	Plastic	1.544	55.9	2.64
5		11.560 (ASP)	0.260				
6	Lens 3	-27.789 (ASP)	0.412	Plastic	1.544	55.9	7.80
7		-3.699 (ASP)	0.244				
8	Lens 4	-0.802 (ASP)	0.284	Plastic	1.634	23.8	-2.68
9		-1.725 (ASP)	0.030				
10	Lens 5	1.609 (ASP)	0.350	Plastic	1.544	55.9	4.16
11		5.126 (ASP)	0.239				
12	Lens 6	2.289 (ASP)	1.170	Plastic	1.544	55.9	-130.32
13		1.818 (ASP)	0.492				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.382				
16	Image	Plano	—				

Reference wavelength (d-line) is 587.6 nm

TABLE 8

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	9.5070E-01	-8.6503E+00	-5.1197E+00	0.0000E+00	0.0000E+00	-1.0237E+01
A4 =	-1.1293E-01	-1.2445E-01	-1.7413E-01	-1.8964E-01	-2.5660E-01	-2.6110E-01
A6 =	2.8449E-01	9.4021E-01	7.9717E-01	-1.2160E-01	-8.5578E-02	-9.5160E-02
A8 =	-4.6548E-01	-1.9379E+00	-1.6134E+00	2.8510E-01	-1.6372E-01	-2.1757E-02
A10 =	3.6510E-01	2.0158E+00	1.5677E+00	-3.3838E-01	3.6027E-01	2.6441E-01
A12 =	-1.5706E-01	-7.8278E-01	-4.4901E-01	-1.0314E-01	2.4226E-01	-9.4393E-02
A14 =				6.5730E-01	-3.0268E-01	-4.5524E-02
Surface #						
	8	9	10	11	12	13
k =	-2.5822E+00	-6.5846E-02	-9.7922E+00	3.0000E+00	-2.0000E+01	-2.3875E+00
A4 =	-3.3405E-01	-2.8652E-01	7.8826E-02	1.7557E-01	-1.9002E-01	-1.8222E-01
A6 =	1.3688E-01	3.2868E-01	-1.9025E-01	-3.8131E-01	3.5324E-02	9.7623E-02
A8 =	4.9723E-01	1.2654E-02	9.2275E-02	3.0878E-01	1.8523E-02	-4.2574E-02
A10 =	-6.8533E-01	-1.1047E-01	-3.0406E-02	-1.7157E-01	-1.0336E-02	1.2934E-02
A12 =	5.3047E-01	4.3996E-02	-9.2850E-04	6.0698E-02	2.2036E-03	-2.4524E-03
A14 =	-1.8280E-01	1.3740E-02	2.3757E-03	-1.1501E-02	-2.3069E-04	2.5592E-04
A16 =				8.6055E-04	9.7129E-06	-1.1161E-05

In the image system lens assembly according to the 4th embodiment, the definitions of f, Fno, HFOV, V4, V5, R11, R12, f1, f2, Σ CT, Td, TTL, and ImgH are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment. Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following relationships:

f (mm)	3.42	f/f1	-0.39
Fno	2.35	f/f2	1.29
HFOV (deg.)	39.4	Σ CT/Td	0.78
V4/V5	0.43	f/tan(HFOV) (mm)	4.16
(R11 - R12)/(R11 + R12)	0.11	TTL/ImgH	1.66

5th Embodiment

FIG. 9 is a schematic view of an image system lens assembly according to the 5th embodiment of the present disclosure. FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 5th embodiment. In FIG. 9, the image system lens assembly includes, in order from an object side to an image side, a first lens element 510, an aperture stop 500, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, an IR-cut filter 580, an image plane 570 and an image sensor 590.

The first lens element 510 with negative refractive power has a convex object-side surface 511 and a concave image-side surface 512. The first lens element 510 is made of plastic

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material and has the object-side surface **511** and the image-side surface **512** being aspheric.

The second lens element **520** with positive refractive power has a convex object-side surface **521** and a convex image-side surface **522**. The second lens element **520** is made of plastic material and has the object-side surface **521** and the image-side surface **522** being aspheric.

The third lens element **530** with positive refractive power has a concave object-side surface **531** and a convex image-side surface **532**. The third lens element **530** is made of plastic material and has the object-side surface **531** and the image-side surface **532** being aspheric.

The fourth lens element **540** with negative refractive power has a concave object-side surface **541** and a convex image-side surface **542**. The fourth lens element **540** is made of plastic material and has the object-side surface **541** and the image-side surface **542** being aspheric.

The fifth lens element **550** with positive refractive power has a convex object-side surface **551** and a concave image-side surface **552**, wherein the object-side surface **551** of the

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fifth lens element **550** changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface **552** of the fifth lens element **550** changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element **550** is made of plastic material and has the object-side surface **551** and the image-side surface **552** being aspheric.

The sixth lens element **560** with positive refractive power has a convex object-side surface **561** and a concave image-side surface **562**, wherein the image-side surface **562** of the sixth lens element **560** changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element **560** is made of plastic material and has the object-side surface **561** and the image-side surface **562** being aspheric.

The IR-cut filter **580** is made of glass material, wherein the IR-cut filter **580** is located between the sixth lens element **560** and the image plane **570**, and will not affect the focal length of the image system lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment							
f = 3.41 mm, Fno = 2.40, HFOV = 39.5 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	2.146 (ASP)	0.212	Plastic	1.640	23.3	-9.28
2		1.515 (ASP)	0.180				
3	Ape. Stop	Plano	-0.116				
4	Lens 2	1.405 (ASP)	0.435	Plastic	1.544	55.9	2.54
5		-80.173 (ASP)	0.319				
6	Lens 3	-8.811 (ASP)	0.389	Plastic	1.544	55.9	17.11
7		-4.597 (ASP)	0.238				
8	Lens 4	-0.851 (ASP)	0.250	Plastic	1.640	23.3	-2.91
9		-1.745 (ASP)	0.030				
10	Lens 5	1.792 (ASP)	0.350	Plastic	1.544	55.9	5.26
11		4.466 (ASP)	0.169				
12	Lens 6	1.702 (ASP)	1.150	Plastic	1.544	55.9	12.63
13		1.724 (ASP)	0.492				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.474				
16	Image	Plano	—				

Reference wavelength (d-line) is 587.6 nm

TABLE 10

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	4.6773E-01	-7.6158E+00	-4.2382E+00	-1.4083E+00	-1.1310E+00	-6.7985E+00
A4 =	-1.2665E-01	-1.3674E-01	-1.7794E-01	-1.7013E-01	-2.8012E-01	-2.6845E-01
A6 =	2.8340E-01	8.7880E-01	7.6429E-01	-4.1169E-02	-6.8685E-02	-1.3804E-01
A8 =	-5.2068E-01	-1.9359E+00	-1.5854E+00	1.4993E-01	-3.8100E-02	3.3581E-02
A10 =	4.0014E-01	2.0133E+00	1.5983E+00	-2.0069E-01	2.4340E-01	2.4922E-01
A12 =	-1.5759E-01	-7.8813E-01	-4.5180E-01	-9.2114E-02	2.4830E-01	-9.5127E-02
A14 =				6.4017E-01	-3.1858E-01	-4.4062E-02
Surface #						
	8	9	10	11	12	13
k =	-2.8534E+00	1.3718E-01	-8.7808E+00	2.9953E+00	-1.1642E+01	-2.4477E+00
A4 =	-3.4035E-01	-2.9922E-01	7.8826E-02	1.7557E-01	-1.9002E-01	-1.8222E-01
A6 =	1.0709E-01	3.4277E-01	-1.9025E-01	-3.8131E-01	3.5324E-02	9.7623E-02
A8 =	4.9964E-01	1.6235E-02	9.2275E-02	3.0878E-01	1.8523E-02	-4.2574E-02

TABLE 10-continued

Aspheric Coefficients						
A10 =	-6.6882E-01	-1.1904E-01	-3.0406E-02	-1.7157E-01	-1.0336E-02	1.2934E-02
A12 =	5.3360E-01	4.6393E-02	-9.2850E-04	6.0698E-02	2.2036E-03	-2.4524E-03
A14 =	-1.8356E-01	1.4629E-02	2.3757E-03	-1.1501E-02	-2.3069E-04	2.5592E-04
A16 =			8.6055E-04	9.7129E-06	-1.1161E-05	

In the image system lens assembly according to the 5th embodiment, the definitions of f , Fno , $HFOV$, $V4$, $V5$, $R11$, $R12$, $f1$, $f2$, ΣCT , Td , TTL , and $ImgH$ are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment. Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following relationships:

f (mm)	3.41	$f/f1$	-0.37
Fno	2.40	$f/f2$	1.29
$HFOV$ (deg.)	39.5	$\Sigma CT/Td$	0.18
$V4/V5$	0.42	$f/\tan(HFOV)$ [mm]	4.16
$(R11 - R12)/(R11 + R12)$	-0.01	$TTL/ImgH$	1.65

6th Embodiment

FIG. 11 is a schematic view of an image system lens assembly according to the 6th embodiment of the present disclosure. FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 6th embodiment. In FIG. 11, the image system lens assembly includes, in order from an object side to an image side, a first lens element 610, an aperture stop 600, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 680, an image plane 670 and an image sensor 690.

The first lens element 610 with negative refractive power has a convex object-side surface 611 and a concave image-side surface 612. The first lens element 610 is made of glass material and has the object-side surface 611 and the image-side surface 612 being aspheric.

The second lens element 620 with positive refractive power has a convex object-side surface 621 and a concave image-

side surface 622. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being aspheric.

The third lens element 630 with positive refractive power has a convex object-side surface 631 and a convex image-side surface 632. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being aspheric.

The fourth lens element 640 with negative refractive power has a concave object-side surface 641 and a convex image-side surface 642. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being aspheric.

The fifth lens element 650 with positive refractive power has a convex object-side surface 651 and a concave image-side surface 652, wherein the object-side surface 651 of the fifth lens element 650 changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface 652 of the fifth lens element 650 changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element 650 is made of plastic material and has the object-side surface 661 and the image-side surface 652 being aspheric.

The sixth lens element 660 with positive refractive power has a convex object-side surface 661 and a concave image-side surface 662, wherein the image-side surface 662 of the sixth lens element 660 changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element 660 is made of plastic material and has the object-side surface 661 and the image-side surface 662 being aspheric.

The IR-cut filter 680 is made of glass material, wherein the IR-cut filter 680 is located between the sixth lens element 660 and the image plane 670, and will not affect the focal length of the image system lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment							
$f = 3.34$ mm, $Fno = 2.35$, $HFOV = 40.1$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	2.348 (ASP)	0.271	Glass	1.566	61.1	-8.97
2		1.538 (ASP)	0.161				
3	Ape. Stop	Plano	-0.088				
4	Lens 2	1.213 (ASP)	0.395	Plastic	1.544	55.9	2.97
5		4.286 (ASP)	0.197				
6	Lens 3	27.942 (ASP)	0.504	Plastic	1.544	55.9	4.67
7		-2.776 (ASP)	0.225				
8	Lens 4	-0.805 (ASP)	0.317	Plastic	1.634	23.8	-2.44
9		-1.937 (ASP)	0.030				
10	Lens 5	1.616 (ASP)	0.350	Plastic	1.544	55.9	4.33
11		4.746 (ASP)	0.261				
12	Lens 6	1.984 (ASP)	1.094	Plastic	1.530	55.8	94.29
13		1.671 (ASP)	0.492				

TABLE 11-continued

6th Embodiment f = 3.34 mm, Fno = 2.35, HFOV = 40.1 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.395				
16	Image	Plano	—				

Reference wavelength (d-line) is 587.6 nm

TABLE 12

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	1.7154E+00	-1.4329E+01	-6.4793E+00	-8.5339E+00	3.0000E+00	-1.0262E+01
A4 =	-1.0363E-01	-1.7215E-01	-2.2472E-01	-2.3906E-01	-2.0831E-01	-2.7040E-01
A6 =	3.2288E-01	9.5846E-01	7.3202E-01	-2.2783E-01	-1.5216E-01	-1.3832E-01
A8 =	-4.9408E-01	-1.8664E+00	-1.6268E+00	3.6413E-01	-1.8549E-01	-2.6490E-02
A10 =	4.2560E-01	1.9931E+00	1.4838E+00	-3.7887E-01	4.6155E-01	3.0963E-01
A12 =	-1.5726E-01	-7.8278E-01	-4.4901E-01	-1.0314E-01	2.4322E-01	-9.3793E-02
A14 =				6.5730E-01	-3.0268E-01	-4.5876E-02
Surface #						
	9	9	10	11	12	13
k =	-2.4866E+00	-1.6398E-01	-9.2136E+00	3.0000E+00	-1.2950E+01	-2.5869E+00
A4 =	-3.5495E-01	-2.8593E-01	7.8826E-02	1.7557E-01	-1.9002E-01	-1.8222E-01
A6 =	1.4097E-01	3.3814E-01	-1.9025E-01	-3.8131E-01	3.5324E-02	9.7623E-02
A8 =	5.3469E-01	1.4654E-02	9.2275E-02	3.0878E-01	1.8523E-02	-4.2574E-02
A10 =	-6.9905E-01	-1.1445E-01	-3.0406E-02	-1.7157E-01	-1.0336E-02	1.2934E-02
A12 =	5.3051E-01	4.0918E-02	-9.2850E-04	6.0698E-02	2.2036E-03	-2.4524E-03
A14 =	-1.8315E-01	9.9301E-03	2.3757E-03	-1.1501E-02	-2.3069E-04	2.5592E-04
A16 =				8.6055E-04	9.7129E-06	-1.1161E-05

In the image system lens assembly according to the 6th embodiment, the definitions of f, Fno, HFOV, V4, V5, R11, R12, f1, f2, Σ CT, Td, TTL, and ImgH are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment. Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following relationships:

f (mm)	3.34	f/f1	-0.37
Fno	2.35	f/f2	1.12
HFOV (deg.)	40.1	Σ CT/Td	0.79
V4/V5	0.43	f/tan(HFOV) (mm)	3.97
(R11 - R12)/(R11 + R12)	0.09	TTL/ImgH	1.66

7th Embodiment

FIG. 13 is a schematic view of an image system lens assembly according to the 7th embodiment of the present disclosure. FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 7th embodiment. In FIG. 13, the image system lens assembly includes, in order from an object side to an image side, a first lens element 710, an aperture stop

700, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, an IR-cut filter 780, an image plane 770 and an image sensor 790.

The first lens element 710 with negative refractive power has a convex object-side surface 711 and a concave image-side surface 712. The first lens element 710 is made of plastic material and has the object-side surface 711 and the image-side surface 712 being aspheric.

The second lens element 720 with positive refractive power has a convex object-side surface 721 and a convex image-side surface 722. The second lens element 720 is made of plastic material and has the object-side surface 721 and the image-side surface 722 being aspheric.

The third lens element 730 with negative refractive power has a concave object-side surface 731 and a convex image-side surface 732. The third lens element 730 is made of plastic material and has the object-side surface 731 and the image-side surface 732 being aspheric.

The fourth lens element 740 with negative refractive power has a concave object-side surface 741 and a convex image-side surface 742. The fourth lens element 740 is made of plastic material and has the object-side surface 741 and the image-side surface 742 being aspheric.

The fifth lens element 750 with positive refractive power has a convex object-side surface 751 and a concave image-side surface 752, wherein the object-side surface 751 of the fifth lens element 750 changes from convex at a paraxial

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region to concave at a peripheral region, and the image-side surface **752** of the fifth lens element **750** changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element **750** is made of plastic material and has the object-side surface **751** and the image-side surface **752** being aspheric.

The sixth lens element **760** with negative refractive power has a convex object-side surface **761** and a concave image-side surface **762**, wherein the image-side surface **762** of the sixth lens element **760** changes from concave at a paraxial

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region to convex at a peripheral region. The sixth lens element **760** is made of plastic material and has the object-side surface **761** and the image-side surface **762** being aspheric.

The IR-cut filter **780** is made of glass material, wherein the IR-cut filter **780** is located between the sixth lens element **760** and the image plane **770**, and will not affect the focal length of the image system lens assembly.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment							
f = 3.68 mm, Fno = 2.50, HFOV = 37.4 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	1.993 (ASP)	0.241	Plastic	1.640	23.3	-7.29
2		1.330 (ASP)	0.184				
3	Ape. Stop	Plano	-0.139				
4	Lens 2	1.366 (ASP)	0.453	Plastic	1.544	55.9	2.33
5		-15.180 (ASP)	0.427				
6	Lens 3	-5.603 (ASP)	0.294	Plastic	1.544	55.9	-59.03
7		-6.912 (ASP)	0.203				
8	Lens 4	-1.079 (ASP)	0.250	Plastic	1.640	23.3	-4.70
9		-1.834 (ASP)	0.039				
10	Lens 5	1.577 (ASP)	0.330	Plastic	1.544	55.9	4.39
11		4.294 (ASP)	0.411				
12	Lens 6	2.705 (ASP)	0.962	Plastic	1.535	56.3	-11.64
13		1.652 (ASP)	0.450				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.495				
16	Image	Plano	—				

Reference wavelength (d-line) is 587.6 nm

TABLE 14

Aspheric Coefficients						
	Surface #					
	1	2	4	5	6	7
k =	-6.6551E-02	-6.5559E+00	-5.0994E+00	-1.7565E+01	1.4593E+00	3.0000E+00
A4 =	-1.3004E-01	-1.2983E-01	-1.6839E-01	-1.1775E-01	-2.9291E-01	-3.4015E-01
A6 =	2.5655E-01	8.7976E-01	8.5828E-01	-2.0722E-02	-6.2942E-03	-2.5717E-02
A8 =	-4.6812E-01	-1.8486E+00	-1.6121E+00	1.0966E-01	-3.7395E-02	-2.8111E-03
A10 =	3.4374E-01	1.7928E+00	1.4785E+00	-8.2521E-02	1.7631E-01	2.4774E-01
A12 =	-1.2206E-01	-7.1223E-01	-4.0300E-01	-1.3566E-01	1.9933E-01	-1.1011E-01
A14 =				4.7317E-01	-2.4101E-01	-3.2864E-02
	Surface #					
	8	9	10	11	12	13
k =	-3.6708E+00	-7.0259E-02	-8.1393E+00	3.0000E+00	-1.9725E+01	-2.9960E+00
A4 =	-3.3483E-01	-2.7649E-01	7.8826E-02	1.7557E-01	-1.9002E-01	-1.8222E-01
A6 =	1.0301E-01	3.0532E-01	-1.9025E-01	-3.8131E-01	3.5324E-02	9.7623E-02
A8 =	4.9124E-01	4.3277E-03	9.2275E-02	3.0878E-01	1.8523E-02	-4.2574E-02
A10 =	-6.8986E-01	-1.1447E-01	-3.0406E-02	-1.7157E-01	-1.0336E-02	1.2934E-02
A12 =	5.3815E-01	5.5969E-02	-9.2850E-04	6.0698E-02	2.2036E-03	-2.4524E-03
A14 =	-1.7648E-01	9.7684E-03	2.3767E-03	-1.1501E-02	-2.3069E-04	2.5592E-04
A16 =				8.6055E-04	9.7129E-06	-1.1161E-05

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In the image system lens assembly according to the 7th embodiment, the definitions of f , Fno , $HFOV$, $V4$, $V5$, $R11$, $R12$, $f1$, $f2$, ΣCT , Td , TTL , and $ImgH$ are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment. Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following relationships:

f (mm)	3.68	$f/f1$	-0.50
Fno	2.50	$f/f2$	1.58
$HFOV$ (deg.)	37.4	$\Sigma CT/Td$	0.69
$V4/V5$	0.42	$f/\tan(HFOV)$ (mm)	4.81
$(R11 - R12)/(R11 + R12)$	0.24	$TTL/ImgH$	1.66

8th Embodiment

FIG. 15 is a schematic view of an image system lens assembly according to the 8th embodiment of the present disclosure. FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 8th embodiment. In FIG. 15, the image system lens assembly includes, in order from an object side to an image side, a first lens element 810, a second lens element 820, an aperture stop 800, a third lens element 830, a fourth lens element 840, a fifth lens element 850, a sixth lens element 860, an IR-cut filter 880, an image plane 870, and an image sensor 890.

The first lens element 810 with negative refractive power has a convex object-side surface 811 and a concave image-side surface 812. The first lens element 810 is made of plastic material and has the object-side surface 811 and the image-side surface 812 being aspheric.

The second lens element 820 with positive refractive power has a convex object-side surface 821 and a convex image-side

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surface 822. The second lens element 820 is made of plastic material and has the object-side surface 821 and the image-side surface 822 being aspheric.

The third lens element 830 with positive refractive power has a concave object-side surface 831 and a convex image-side surface 832. The third lens element 830 is made of plastic material and has the object-side surface 831 and the image-side surface 832 being aspheric.

The fourth lens element 840 with negative refractive power has a concave object-side surface 841 and a convex image-side surface 842. The fourth lens element 840 is made of plastic material and has the object-side surface 841 and the image-side surface 842 being aspheric.

The fifth lens element 850 with positive refractive power has a convex object-side surface 851 and a concave image-side surface 852, wherein the object-side surface 851 of the fifth lens element 850 changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface 852 of the fifth lens element 850 changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element 850 is made of plastic material and has the object-side surface 851 and the image-side surface 852 being aspheric.

The sixth lens element 860 with negative refractive power has a convex object-side surface 861 and a concave image-side surface 862, wherein the image-side surface 862 of the sixth lens element 860 changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element 860 is made of plastic material and has the object-side surface 861 and the image-side surface 862 being aspheric.

The IR-cut filter 880 is made of glass material, wherein the IR-cut filter 880 is located between the sixth lens element 860 and the image plane 870, and will not affect the focal length of the image system lens assembly.

The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment							
$f = 3.39$ mm, $Fno = 2.20$, $HFOV = 38.5$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	2.166 (ASP)	0.153	Plastic	1.640	23.3	-10.02
2		1.575 (ASP)	0.045				
3	Lens 2	1.473 (ASP)	0.538	Plastic	1.544	55.9	2.67
4		-100.000 (ASP)	-0.001				
5	Ape. Stop	Plano	0.416				
6	Lens 3	-6.478 (ASP)	0.400	Plastic	1.544	55.9	17.31
7		-3.922 (ASP)	0.275				
8	Lens 4	-1.007 (ASP)	0.250	Plastic	1.650	21.4	-4.11
9		-1.775 (ASP)	0.030				
10	Lens 5	1.814 (ASP)	0.507	Plastic	1.544	55.9	3.82
11		12.906 (ASP)	0.316				
12	Lens 6	2.617 (ASP)	0.793	Plastic	1.530	55.8	-6.61
13		1.341 (ASP)	0.450				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.282				
16	Image	Plano	—				

Reference wavelength (d-line) is 587.6 nm

TABLE 16

Aspheric Coefficients						
Surface #						
	1	2	3	4	6	7
k =	4.4038E-02	-7.8141E+00	-3.3795E+00	-1.0000E+00	-1.0000E+00	-7.6897E+00
A4 =	-1.2905E-01	-1.1984E-01	-1.5587E-01	-6.9591E-02	-2.2264E-01	-2.9389E-01
A6 =	2.9857E-01	9.0811E-01	8.4778E-01	-5.8127E-02	-2.6840E-02	-2.9731E-02
A8 =	-5.1595E-01	-1.8798E+00	-1.5275E+00	1.0845E-01	-1.4853E-01	-8.0225E-02
A10 =	3.5499E-01	1.7643E+00	1.3745E+00	1.8344E-01	1.2817E-01	2.1444E-01
A12 =	-1.0179E-01	-6.6164E-01	-4.0218E-01	-5.1178E-01	1.6134E-01	-9.6216E-02
A14 =				4.7318E-01	-1.1844E-01	1.3470E-02
Surface #						
	8	9	10	11	12	13
k =	-3.4056E+00	1.8392E-02	-1.5100E+01	2.1574E+00	-1.1532E+01	-3.2940E+00
A4 =	-3.5926E-01	-2.6585E-01	7.6862E-02	2.2322E-01	-1.8406E-01	-1.6246E-01
A6 =	9.8381E-02	3.0181E-01	-1.5997E-01	-3.6728E-01	3.3614E-02	9.2535E-02
A8 =	4.8811E-01	-3.1396E-03	9.0526E-02	3.0769E-01	1.8599E-02	-4.2289E-02
A10 =	-7.0095E-01	-1.1924E-01	-2.2593E-02	-1.7255E-01	-1.0054E-02	1.2991E-02
A12 =	5.1312E-01	5.3016E-02	-9.9750E-04	6.0487E-02	2.2845E-03	-2.4528E-03
A14 =	-1.4732E-01	6.8630E-03	2.4837E-03	-1.1517E-02	-2.3372E-04	2.5544E-04
A16 =			-9.6081E-05	8.7603E-04	-9.4301E-06	-1.1294E-05

In the image system lens assembly according to the 8th embodiment, the definitions of f, Fno, HFOV, V4, V5, R11, R12, f1, f2, Σ CT, Td, TTL, and ImgH are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment. Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following relationships:

f (mm)	3.39	f/f1	-0.34
Fno	2.20	f/f2	1.27
HFOV (deg.)	38.5	Σ CT/Td	0.71
V4/V5	0.38	f/tan(HFOV) (mm)	4.27
(R11 - R12)/(R11 + R12)	0.32	TTL/ImgH	1.67

9th Embodiment

FIG. 17 is a schematic view of an image system lens assembly according to the 9th embodiment of the present disclosure. FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 9th embodiment. In FIG. 17, the image system lens assembly includes, in order from an object side to an image side, a first lens element 910, an aperture stop 900, a second lens element 920, a third lens element 930, a fourth lens element 940, a fifth lens element 950, a sixth lens element 960, an IR-cut filter 980, an image plane 970 and an image sensor 990.

The first lens element 910 with negative refractive power has a convex object-side surface 911 and a concave image-side surface 912. The first lens element 910 is made of plastic material and has the object-side surface 911 and the image-side surface 912 being aspheric.

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The second lens element 920 with positive refractive power has a convex object-side surface 921 and a planar image-side surface 922. The second lens element 920 is made of glass material and has the object-side surface 921 and the image-side surface 922 being aspheric.

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The third lens element 930 with positive refractive power has a concave object-side surface 931 and a convex image-side surface 932. The third lens element 930 is made of plastic material and has the object-side surface 931 and the image-side surface 932 being aspheric.

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The fourth lens element 940 with negative refractive power has a concave object-side surface 941 and a convex image-side surface 942. The fourth lens element 940 is made of plastic material and has the object-side surface 941 and the image-side surface 942 being aspheric.

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The fifth lens element 950 with positive refractive power has a convex object-side surface 951 and a convex image-side surface 952, wherein the object-side surface 951 of the fifth lens element 950 changes from convex at a paraxial region to concave at a peripheral region. The fifth lens element 950 is made of plastic material and has the object-side surface 951 and the image-side surface 952 being aspheric.

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The sixth lens element 960 with negative refractive power has a convex object-side surface 961 and a concave image-side surface 962, wherein the image-side surface 962 of the sixth lens element 960 changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element 960 is made of plastic material and has the object-side surface 961 and the image-side surface 962 being aspheric.

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The IR-cut filter 980 is made of glass material, wherein the IR-cut filter 980 is located between the sixth lens element 960 and the image plane 970, and will not affect the focal length of the image system lens assembly.

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The detailed optical data of the 9th embodiment are shown in Table 17 and the aspheric surface data are shown in Table 18 below.

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TABLE 17

9th Embodiment f = 3.20 mm, Fno = 2.15, HFOV = 40.7 deg.							
Surface #		Curvature Radius	thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	2.015 (ASP)	0.250	Plastic	1.640	23.3	-19.53
2		1.652 (ASP)	0.189				
3	Ape. Stop	Plano	-0.130				
4	Lens 2	1.654 (ASP)	0.440	Glass	1.542	62.9	3.05
5		∞ (ASP)	0.287				
6	Lens 3	-14.818 (ASP)	0.391	Plastic	1.535	56.3	11.27
7		-4.322 (ASP)	0.285				
8	Lens 4	-0.905 (ASP)	0.279	Plastic	1.640	23.3	-3.27
9		-1.790 (ASP)	0.030				
10	Lens 5	2.157 (ASP)	0.500	Plastic	1.535	56.3	3.90
11		-59.694 (ASP)	0.216				
12	Lens 6	1.620 (ASP)	0.754	Plastic	1.535	56.3	-19.76
13		1.177 (ASP)	0.450				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.408				
16	Image	Plano	—				

Reference wavelength (d-line) is 587.6 nm

TABLE 18

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	5.2664E-01	-5.9283E+00	-3.5530E+00	0.0000E+00	0.0000E+00	-1.9743E+01
A4 =	-1.1422E-01	-1.1773E-01	-1.5036E-01	-1.5013E-01	-2.6119E-01	-2.8064E-01
A6 =	2.5456E-01	8.7307E-01	8.3271E-01	-3.9042E-02	-4.5031E-02	-6.4552E-02
A8 =	-5.2149E-01	-1.8891E+00	-1.7372E+00	2.1926E-01	-9.7777E-02	-3.1606E-02
A10 =	4.1126E-01	1.8307E+00	1.8507E+00	-2.7683E-01	1.6233E-01	2.5031E-01
A12 =	-1.5035E-01	-6.7950E-01	-7.0115E-01	-1.8952E-01	2.2136E-01	-9.5452E-02
A14 =				6.5005E-01	-1.7167E-01	-3.2762E-02
Surface #						
	8	9	10	11	12	13
k =	-2.6923E+00	4.0053E-01	-1.6497E+01	-2.0000E+01	-5.0080E+00	-2.8660E+00
A4 =	-3.7062E-01	-2.8757E-01	7.7593E-02	2.3871E-01	-2.0185E-01	-1.7479E-01
A6 =	9.2214E-02	3.0050E-01	-1.5090E-01	-3.6866E-01	3.1090E-02	9.7308E-02
A8 =	4.8736E-01	1.8422E-03	8.8800E-02	3.0707E-01	1.8593E-02	-4.2893E-02
A10 =	-6.9313E-01	-1.1638E-01	-3.2254E-02	-1.7296E-01	-9.9549E-03	1.2945E-02
A12 =	5.3321E-01	5.3399E-02	-8.6395E-04	6.0413E-02	2.2881E-03	-2.4451E-03
A14 =	-1.6590E-01	8.0930E-03	2.0994E-03	-1.1504E-02	-2.3623E-04	2.5696E-04
A16 =				8.9606E-04	-7.1230E-06	-1.1525E-05

In the image system lens assembly according to the 9th embodiment, the definitions of f, Fno, HFOV, V4, V5, R11, R12, f1, f2, Σ CT, Td, TTL, and ImgH are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment. Moreover, these parameters can be calculated from Table 17 and Table 18 as the following values and satisfy the following relationships:

f (mm)	3.20	f/f1	-0.16
Fno	2.15	f/f2	1.05
HFOV (deg.)	40.7	Σ CT/Td	0.75
V4/V5	0.41	f/tan(HFOV) (mm)	3.72
(R11 - R12)/(R11 + R12)	0.16	TTL/ImgH	1.60

10th Embodiment

FIG. 19 is a schematic view of an image system lens assembly according to the 10th embodiment of the present disclosure. FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image system lens assembly according to the 10th embodiment. In FIG. 19, the image system lens assembly includes, in order from an object side to an image side, a first lens element 1010, aperture stop 1000, a second lens element 1020, a third lens element 1030, a fourth lens element 1040, a fifth lens element 1050, a sixth lens element 1060, an IR-cut filter 1080, an image plane 1070, and an image sensor 1090.

The first lens element 1010 with negative refractive power has a convex object-side surface 1011 and a concave image-side surface 1012. The first lens element 1010 is made of

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plastic material and has the object-side surface **1011** and the image-side surface **1012** being aspheric.

The second lens element **1020** with positive refractive power has a convex object-side surface **1021** and a concave image-side surface **1022**. The second lens element **1020** is made of plastic material and has the object-side surface **1021** and the image-side surface **1022** being aspheric.

The third lens element **1030** with positive refractive power has a convex object-side surface **1031** and a convex image-side surface **1032**. The third lens element **1030** is made of plastic material and has the object-side surface **1031** and the image-side surface **1032** being aspheric.

The fourth lens element **1040** with negative refractive power has a concave object-side surface **1041** and a convex image-side surface **1042**. The fourth lens element **1040** is made of plastic material and has the object-side surface **1041** and the image-side surface **1042** being aspheric.

The fifth lens element **1050** with positive refractive power has a convex object-side surface **1051** and a concave image-side surface **1052**, wherein the object-side surface **1051** of the

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fifth lens element **1050** changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface **1052** of the fifth lens element **1050** changes from concave at a paraxial region to convex at a peripheral region. The fifth lens element **1050** is made of plastic material and has the object-side surface **1051** and the image-side surface **1052** being aspheric.

The sixth lens element **1060** with positive refractive power has a convex object-side surface **1061** and a concave image-side surface **1062**, wherein the image-side surface **1062** of the sixth lens element **1060** changes from concave at a paraxial region to convex at a peripheral region. The sixth lens element **1060** is made of plastic material and has the object-side surface **1061** and the image-side surface **1062** being aspheric.

The IR-cut filter **1080** is made of glass material, wherein the IR-cut filter **1080** is located between the sixth lens element **1060** and the image plane **1070**, and will not affect the focal length of the image system lens assembly.

The detailed optical data of the 10th embodiment are shown in Table 19 and the aspheric surface data are shown in Table 20 below.

TABLE 19

10th Embodiment							
f = 3.20 mm, Fno = 2.10, HFOV = 41.2 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	2.646 (ASP)	0.250	Plastic	1.633	23.4	-50.39
2		2.353 (ASP)	0.165				
3	Ape. Stop	Plano	-0.125				
4	Lens 2	1.825 (ASP)	0.493	Plastic	1.535	56.3	3.99
5		11.517 (ASP)	0.217				
6	Lens 3	22.861 (ASP)	0.422	Plastic	1.535	56.3	4.89
7		-2.932 (ASP)	0.283				
8	Lens 4	-0.814 (ASP)	0.395	Plastic	1.640	23.3	-2.51
9		-1.968 (ASP)	0.030				
10	Lens 5	2.157 (ASP)	0.518	Plastic	1.535	56.3	4.37
11		26.055 (ASP)	0.171				
12	Lens 6	1.402 (ASP)	0.751	Plastic	1.535	56.3	42.85
13		1.214 (ASP)	0.450				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.503				
16	Image	Plano	—				

Reference wavelength (d-line) is 587.6 nm

TABLE 20

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	2.8340E+00	-4.8095E+00	-1.8680E+00	0.0000E+00	0.0000E+00	-1.5668E+01
A4 =	-7.0480E-02	-1.0015E-01	-1.3063E-01	-1.9429E-01	-2.6177E-01	-2.9526E-01
A6 =	2.2492E-01	8.9371E-01	7.9423E-01	-9.3708E-02	-1.2458E-01	-1.1239E-01
A8 =	-5.2497E-01	-1.9763E+00	-1.7299E+00	1.5989E-01	-2.2011E-01	-5.7691E-02
A10 =	4.5921E-01	1.7809E+00	1.6046E+00	-2.0472E-01	4.3030E-02	2.6240E-01
A12 =	-1.5691E-01	-4.1543E-01	-5.4239E-01	-2.5973E-01	2.6220E-01	-8.1769E-02
A14 =				3.5845E-01	-4.3999E-02	-6.8450E-03
Surface #						
	8	9	10	11	12	13
k =	-2.3292E+00	5.5512E-01	-1.4135E+01	-1.0000E+00	-4.2043E+00	-2.5566E+00
A4 =	-3.8380E-01	-2.8622E-01	7.1440E-02	2.4203E-01	-1.8277E-01	-1.8489E-01
A6 =	9.6448E-02	2.8447E-01	-1.3602E-01	-3.6245E-01	3.0499E-02	1.0128E-01
A8 =	4.9760E-01	-5.4578E-05	8.2955E-02	3.0668E-01	1.8059E-02	-4.3474E-02

TABLE 20-continued

Aspheric Coefficients						
A10 =	-6.8021E-01	-1.1338E-01	-3.1862E-02	-1.7315E-01	-9.9588E-03	1.2920E-02
A12 =	5.4383E-01	5.3163E-02	6.1060E-05	6.0409E-02	2.2856E-03	-2.4295E-03
A14 =	-1.8215E-01	3.2618E-03	1.9378E-03	-1.1492E-02	-2.3351E-04	2.5720E-04
A16 =				8.9578E-04	-2.5947E-06	-1.1701E-05

In the image system lens assembly according to the 10th embodiment, the definitions of f , Fno , $HFOV$, $V4$, $V5$, $R11$, $R12$, $f1$, $f2$, ΣCT , Td , TTL , and $ImgH$ are the same as those stated in the 1st embodiment with corresponding values for the 10th embodiment. Moreover, these parameters can be calculated from Table 19 and Table 20 as the following values and satisfy the following relationships:

f (mm)	3.20	$f/f1$	-0.06
Fno	2.10	$f/f2$	0.80
$HFOV$ (deg.)	41.2	$\Sigma CT/Td$	0.79
$V4/V5$	0.41	$f/\tan(HFOV)$ (mm)	3.66
$(R11 - R12)/(R11 + R12)$	0.07	$TTL/ImgH$	1.63

It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims.

What is claimed is:

1. An image system lens assembly comprising, in order from an object side to an image side:

a first lens element with negative refractive power having a convex object-side surface and a concave image-side surface;

a second lens element with positive refractive power;

a third lens element with refractive power;

a fourth lens element with negative refractive power having a concave object-side surface and a convex image-side surface;

a fifth lens element with positive refractive power made of plastic material, wherein at least one of an object-side surface and an image-side surface of the fifth lens element is aspheric; and

a sixth lens element with refractive power made of plastic material, and having a convex object-side surface and a concave image-side surface, wherein the image-side surface of the sixth lens element changes from concave at a paraxial region to convex at a peripheral region, and at least one of the object-side surface and the image-side surface of the sixth lens element is aspheric;

wherein a focal length of the image system lens assembly is f , a focal length of the first lens element is $f1$, a focal length of the second lens element is $f2$, and the following relationships are satisfied:

$$-0.8 < f/f1 < 0; \text{ and}$$

$$0.7 < f/f2 < 2.4.$$

2. The image system lens assembly of claim 1, wherein an object-side surface of the second lens element and the object-side surface of the fifth lens element are convex.

3. The image system lens assembly of claim 2, wherein the focal length of the image system lens assembly is f , the focal length of the first lens element is $f1$, the focal length of the

second lens element is $f2$, a focal length of the third lens element is $f3$, a focal length of the fourth lens element is $f4$, a focal length of the fifth lens element is $f5$, a focal length of the sixth lens element is $f6$, and the following relationship is satisfied:

$$f/f2 > f/fi, i=1,3,4,5,6.$$

4. The image system lens assembly of claim 3, wherein an axial distance between the first lens element and the second lens element is $T12$, an axial distance between the second lens element and the third lens element is $T23$, an axial distance between the third lens element and the fourth lens element is $T34$, an axial distance between the fourth lens element and the fifth lens element is $T45$, an axial distance between the fifth lens element and the sixth lens element is $T56$, and the following relationships are satisfied:

$$T23 > T12;$$

$$T23 > T34;$$

$$T23 > T45; \text{ and}$$

$$T23 > T56.$$

5. The image system lens assembly of claim 3, wherein an Abbe number of the fourth lens element is $V4$, an Abbe number of the fifth lens element is $V5$, and the following relationship is satisfied:

$$0.2 < V4/V5 < 0.6.$$

6. The image system lens assembly of claim 3, wherein the focal length of the image system lens assembly is f , a half of a maximal field of view of the image system lens assembly is $HFOV$, and the following relationship is satisfied:

$$3.0 \text{ mm} < f/\tan(HFOV) < 6.0 \text{ mm}.$$

7. The image system lens assembly of claim 2, wherein the image-side surface of the fifth lens element is concave.

8. The image system lens assembly of claim 2, wherein the object-side surface of the fifth lens element changes from convex at a paraxial region to concave at a peripheral region.

9. The image system lens assembly of claim 8, wherein a curvature radius of the object-side surface of the sixth lens element is $R11$, a curvature radius of the image-side surface of the sixth lens element is $R12$, and the following relationship is satisfied:

$$-0.10 < (R11 - R12)/(R11 + R12) < 0.45.$$

10. The image system lens assembly of claim 8, wherein the focal length of the image system lens assembly is f , the focal length of the second lens element is $f2$, and the following relationship is satisfied:

$$1.0 < f/f2 < 1.8.$$

11. The image system lens assembly of claim 1, wherein a sum of central thicknesses of the first through sixth lens elements is ΣCT , an axial distance between the object-side

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surface of the first lens element and the image-side surface of the sixth lens element is T_d , and the following relationship is satisfied:

$$0.65 < \Sigma CT / T_d < 0.90.$$

12. The image system lens assembly of claim 1, wherein a maximum image height of the image system lens assembly is $ImgH$, an axial distance between the object-side surface of the first lens element and an image plane is TTL , and the following relationship is satisfied:

$$TTL / ImgH < 1.8.$$

13. An image system lens assembly comprising, in order from an object side to an image side:

a first lens element with negative refractive power having a convex object-side surface and a concave image-side surface;

a second lens element with positive refractive power;

a third lens element with refractive power;

a fourth lens element with negative refractive power;

a fifth lens element with positive refractive power made of plastic material, wherein at least one of an object-side surface and an image-side surface of the fifth lens element is aspheric; and

a sixth lens element with refractive power made of plastic material, and having a concave image-side surface, wherein the image-side surface of the sixth lens element changes from concave at a paraxial region to convex at a peripheral region, and at least one of an object-side surface and the image-side surface of the sixth lens element is aspheric;

wherein a focal length of the image system lens assembly is f , a focal length of the second lens element is f_2 , a sum of central thicknesses of the first through sixth lens elements is ΣCT , an axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is T_d , and the following relationships are satisfied:

$$0.55 < \Sigma CT / T_d < 0.90; \text{ and}$$

$$0.7 < f / f_2 < 2.4.$$

14. The image system lens assembly of claim 13, wherein an object-side surface of the second lens element and the object-side surface of the fifth lens element are convex.

15. The image system lens assembly of claim 14, wherein the sum of central thicknesses of the first through sixth lens elements is ΣCT , the axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is T_d , and the following relationship is satisfied:

$$0.65 < \Sigma CT / T_d < 0.90.$$

16. The image system lens assembly of claim 15, wherein an Abbe number of the fourth lens element is V_4 , an Abbe number of the fifth lens element is V_5 , and the following relationship is satisfied:

$$0.20 < V_4 / V_5 < 0.60.$$

17. The image system lens assembly of claim 15, wherein the focal length of the image system lens assembly is f , a half of a maximal field of view of the image system lens assembly is $HFOV$, and the following relationship is satisfied:

$$3.0 \text{ mm} < \tan(HFOV) < 6.0 \text{ mm}.$$

18. The image system lens assembly of claim 14, wherein the image-side surface of the fifth lens element is concave.

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19. The image system lens assembly of claim 18, wherein an object-side surface of the fourth lens element is concave, and an image-side surface of the fourth lens element is convex.

20. The image system lens assembly of claim 18, wherein the object-side surface of the fifth lens element changes from convex at a paraxial region to concave at a peripheral region, and the image-side surface of the fifth lens element changes from concave at a paraxial region to convex at a peripheral region.

21. The image system lens assembly of claim 18, wherein an axial distance between the first lens element and the second lens element is T_{12} , an axial distance between the second lens element and the third lens element is T_{23} , an axial distance between the third lens element and the fourth lens element is T_{34} , an axial distance between the fourth lens element and the fifth lens element is T_{45} , an axial distance between the fifth lens element and the sixth lens element is T_{56} , and the following relationships are satisfied:

$$T_{23} > T_{12};$$

$$T_{23} > T_{34};$$

$$T_{23} > T_{45}; \text{ and}$$

$$T_{23} > T_{56}.$$

22. The image system lens assembly of claim 13, wherein a maximum image height of the image system lens assembly is $ImgH$, an axial distance between the object-side surface of the first lens element and an image plane is TTL , and the following relationship is satisfied:

$$TTL / ImgH < 1.8.$$

23. An image system lens assembly comprising, in order from an object side to an image side:

a first lens element with negative refractive power having a convex object-side surface and a concave image-side surface;

a second lens element with positive refractive power;

a third lens element with refractive power;

a fourth lens element with negative refractive power;

a fifth lens element with positive refractive power made of plastic material, wherein at least one of an object-side surface and an image-side surface of the fifth lens element is aspheric; and

a sixth lens element with refractive power made of plastic material, and having a convex object-side surface and a concave image-side surface, wherein the image-side surface of the sixth lens element changes from concave at a paraxial region to convex at a peripheral region, and at least one of the object-side surface and the image-side surface of the sixth lens element is aspheric;

wherein a focal length of the image system lens assembly is f , a focal length of the second lens element is f_2 , a half of a maximal field of view of the image system lens assembly is $HFOV$, and the following relationships are satisfied:

$$3.0 \text{ mm} < f / \tan(HFOV) < 6.0 \text{ mm}; \text{ and}$$

$$0.7 < f / f_2 < 2.4.$$

24. The image system lens assembly of claim 23, wherein the object-side surface of the fifth lens element is convex, and the image-side surface of the fifth lens element is concave.

25. The image system lens assembly of claim 24, wherein the focal length of the image system lens assembly is f , the

focal length of the image system lens assembly is f2, and the following relationship is satisfied:

$$1.0 < f/f2 < 1.8.$$

26. The image system lens assembly of claim 23, wherein an object-side surface of the fourth lens element is concave, and an image-side surface of the fourth lens element is convex.

27. The image system lens assembly of claim 23, wherein a maximum image height of the image system lens assembly is ImgH, an axial distance between the object-side surface of the first lens element and an image plane is TTL, and the following relationship is satisfied:

$$TTL/ImgH < 1.8.$$

28. The image system lens assembly of claim 23, wherein the focal length of the image system lens assembly is f, a focal length of the first lens element is f1, the focal length of the second lens element is f2, a focal length of the third lens element is f3, a focal length of the fourth lens element is f4, a focal length of the fifth lens element is f5, a focal length of the sixth lens element is f6, and the following relationship is satisfied:

$$f/f2 > f/fi, i=1,3,4,5,6.$$

29. The image system lens assembly of claim 23, wherein a sum of central thicknesses of the first through sixth lens elements is ΣCT, an axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is Td, and the following relationship is satisfied:

$$0.65 < \Sigma CT/Td < 0.90.$$

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 8,687,292 B2
APPLICATION NO. : 13/665106
DATED : April 1, 2014
INVENTOR(S) : Tsung-Han Tsai and Ming-Ta Chou

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

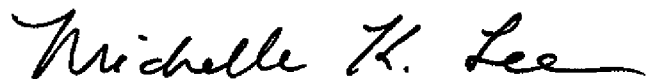
On the Title Page

Item (71) of the issued patent, the name of Applicant should be changed to “Largan Precision Co., Ltd., Taichung (TW)”.

In the Claims

1. In column 35, line 65, claim 3 of the issued patent, “the focal length the image system lens assembly is f ” should be changed to “the focal length of the image system lens assembly is f ”; and “a focal length of the fourth lens element if f_4 ” should be changed to “a focal length of the fourth lens element is f_4 ”.
2. In column 37, line 64, claim 17 of the issued patent, “ $3.0\text{ mm} < \tan(\text{HFOV}) < 6.0\text{ mm}$ ” should be changed to “ $3.0\text{ mm} < f/\tan(\text{HFOV}) < 6.0\text{ mm}$ ”.
3. In column 39, line 1, claim 25 of the issued patent, “the focal length of the image system lens assembly is f_2 ” should be changed to “the focal length of the second lens element is f_2 ”.
4. In column 39, line 19, claim 28 of the issued patent, “a focal length of the fourth lens element if f_4 ” should be changed to “a focal length of the fourth lens element is f_4 ”.

Signed and Sealed this
Eighth Day of July, 2014



Michelle K. Lee
Deputy Director of the United States Patent and Trademark Office